

Máster en Física Avanzada

Especialidad Física Teórica



Trabajo Fin de Máster

Fast Simulation of Neutrino Oscillations in Matter

Juan Ramón Díaz Santos

Tutor: Roberto Ruiz de Austri

Curso académico 2025/26

Abstract

This Master’s Thesis presents **tpeanuts**, a modernisation of the PEANUTS neutrino-oscillation toolkit aimed at fast, differentiable and accelerator-friendly neutrino simulation, extended into a full detector-level statistical-inference framework. The project translates the numerical core of PEANUTS from an array-based Python workflow into a PyTorch implementation while preserving its physical conventions for three-flavour oscillations, matter effects and solar-neutrino propagation, and then builds on top of it a forward-folding detector layer and a gradient-based inference engine that connect oscillation probabilities directly to published experimental data.

The work reviews the theoretical basis of neutrino mixing, vacuum oscillations, the Mikheyev–Smirnov–Wolfenstein matter effect, non-standard interactions, sterile neutrinos and the statistical-inference formalism (Poisson and Gaussian likelihoods, gradient-based fitting, Laplace/Fisher uncertainties) used throughout the analyses. It then describes the software strategy followed in **tpeanuts**: identification of the relevant PEANUTS modules, translation of numerical operations to tensor operations, preservation of units and conventions, construction of validation tests against the original implementation and against **nuSQuIDS**, and preparation of benchmarks for CPU and GPU execution.

The code is organised around a modular architecture in which solar, atmospheric and matter models, new-physics scenarios, detector responses and inference procedures can be selected and combined rather than hard-coded into a single implementation. Dedicated modules support beyond-the-Standard-Model physics, including non-standard interactions and sterile neutrinos, and a generative diffusion model synthesises differential atmospheric-neutrino flux structures. Built on top of this engine, a detector layer folds oscillated fluxes into predicted event counts for five real experiments (Borexino, Daya Bay, IceCube DeepCore, and SNO phases I and II), and an inference layer fits oscillation and detector parameters to their real, published data via automatic-differentiation-based likelihood minimisation. The most complete case, Daya Bay, is validated against the collaboration’s own published results: a near/far ratio fit recovers both $\sin^2 2\theta_{13}$ and Δm_{31}^2 jointly, within 1–2% of Daya Bay’s published values. This establishes **tpeanuts** as a working foundation for real, gradient-based neutrino-oscillation data analysis, not only for fast forward simulation.

Keywords

neutrino oscillations; PEANUTS; PyTorch; fast simulation; MSW effect; solar neutrinos; reactor antineutrinos; statistical inference; diffusion models; atmospheric neutrinos.

Contents

Abstract	I
1 Introduction	1
1.1 Motivation	1
1.2 Objectives	2
1.3 Document structure	3
2 Theoretical Framework	3
2.1 Neutrino propagation fundamentals	3
2.1.1 Neutrino mixing	3
2.1.2 Vacuum oscillations	4
2.1.3 Matter effects	5
2.2 Neutrino sources	6
2.2.1 Solar neutrinos	7
2.2.2 Atmospheric neutrinos	8
2.2.3 Reactor antineutrinos	8
2.2.4 Other neutrino sources	8
2.3 New physics	9
2.3.1 Non-standard interactions (NSI)	9
2.3.2 Sterile neutrinos	10
2.4 Statistical inference formalism	11
2.5 Diffusion generative model	12
2.6 Related computational tools	13
3 Methodology	13
3.1 Translation methodology	14
3.1.1 Analysis and translation	14
3.1.2 Verification of units and physical conventions	14
3.2 Validation strategy	14
3.2.1 Comparison with PEANUTS and nuSQuIDS	14
3.3 Reproduction of experimental and theoretical results	15
3.4 Benchmarking	16
3.4.1 Benchmarking procedure and CPU/GPU performance	17
3.5 Detector-level validation	17
4 Technical Development	18
4.1 Software architecture	18
4.1.1 Configuration layer (<code>config/</code>)	19

4.1.2	Core layer (core/)	19
4.1.3	Medium-specific physics (medium/)	19
4.1.4	Source layer (source/)	20
4.1.5	External-tool integrations (external/)	20
4.1.6	Detector layer (detector/)	20
4.1.7	Inference layer (inference/)	20
4.1.8	Pipeline and orchestration (pipeline/)	21
4.1.9	Utilities (util/)	21
4.1.10	Validation and performance methodology	22
4.2	Numerical simulation conventions	22
4.2.1	Non-dimensionalisation of the Hamiltonian	22
4.2.2	Neutrino/antineutrino convention	23
4.2.3	Precision and evaluation mode	23
4.3	Neutrino detection pipelines	23
4.3.1	Solar-neutrino detection pipeline	24
4.3.2	Atmospheric-neutrino detection pipeline	25
4.4	Test suite and sanity checks	25
4.4.1	Code consistency checks	25
4.4.2	Mathematical checks	26
4.4.3	Physical coherence checks	26
4.4.4	Detector and inference validation	26
4.4.5	Notebook-level validation	26
5	Validation and Results	26
5.1	Validation	27
5.1.1	tpeanuts vs PEANUTS	27
5.1.2	tpeanuts vs nuSQuIDS	27
5.1.3	Internal validation	28
5.2	Benchmark	28
5.2.1	tpeanuts vs PEANUTS	28
5.2.2	tpeanuts vs nuSQuIDS	29
5.3	Pipeline analysis	30
5.3.1	Solar neutrino	30
5.3.2	Atmospheric neutrino	31
5.4	Reproduction of experimental and theoretical results	32
5.4.1	Solar day–night asymmetry	32
5.4.2	MSW resonance	32
5.4.3	Adiabatic approximation	33

5.4.4	CP violation and CPT symmetry	34
5.4.5	Atmospheric neutrino oscillograms	34
5.4.6	Earth matter tomography	36
5.4.7	LMA-Dark degeneracy	36
5.5	BSM extensions	37
5.5.1	NSI	37
5.5.2	Sterile neutrino	37
5.6	Detector-level case study: Daya Bay	38
5.7	Other detector-inference applications	40
6	Conclusions	40
A	Supplementary Material	44

List of Figures

1	Maximum absolute and relative differences between tpeanuts and legacy PEANUTS, by validated module (solar, Earth, vacuum, pipeline). Dashed and dotted lines mark the 1 ppm and 1 ppb reference levels.	27
2	Maximum absolute and relative differences between tpeanuts and nuSQuIDS , by validated configuration (vacuum, atmosphere, solar, solar-Earth-detection pipeline). Dashed and dotted lines mark the 1 ppm and 1 ppb reference levels.	28
3	Wall-clock speed-up of tpeanuts over legacy PEANUTS for the Earth-propagation probability, as a function of energy-grid and nadir-grid size.	29
4	Wall-clock speed-up of tpeanuts over nuSQuIDS for the integrated solar-detector probability, as a function of energy-grid and nadir-grid size.	30
5	Total solar-neutrino flux at the detector, $\Phi_{\alpha}^{\text{tot}}(E, \eta_{\text{fix}})$, summed over all solar production channels, at fixed nadir angle $\eta_{\text{fix}} = 0.17\pi$	30
6	ν_{μ} production-height probability density, $f(h E, \alpha)$, at fixed zenith angle $\alpha = 31.8^{\circ}$, generated with MCEq (top) (Fedynitch et al., 2015) and with the Honda/HKKM parametrisation (bottom) (Honda et al., 2007).	31
7	Height-averaged, detector-level ν_{μ} survival probability $P_{\text{det}}(\nu_{\mu} \rightarrow \nu_{\mu}; E, \theta)$, combining atmospheric and Earth propagation.	32
8	Flux-weighted, energy-integrated day–night asymmetry $A_{\text{DN}}^{\text{int}}(E > 3.5 \text{ MeV})$ for four representative detector sites, compared with the Super-Kamiokande-IV measurement (Abe et al., 2016).	33
9	Local survival probability $P_{ee}(\rho, E)$ across the solar production region (left), and radial cross-sections at fixed energy (right), showing the transition between the adiabatic matter-eigenstate limit and the vacuum-averaged value.	33
10	Appearance probabilities $P_{\mu e}$ and $\bar{P}_{\mu e}$ at a DUNE-like baseline (top) and the resulting CP asymmetry A_{CP} (bottom), for the best-fit and maximal-CP-violation values of δ_{CP}	34
11	Atmospheric $P(\nu_{\mu} \rightarrow \nu_{\mu})$ oscillogram in the $(\cos \theta_z, E)$ plane. The dashed line marks the analytic first vacuum-oscillation minimum, $1.267 \Delta m_{3\ell}^2 L/E = \pi/2$; the red marker shows the nearest grid point.	35
12	Day–night asymmetry versus energy at fixed nadir angle ($\eta = 30^{\circ}$, 1 km depth), comparing the standard LMA solution, a sign-flip demonstration with an Earth-only NSI coupling, and the full LMA-Dark approximation, against the Super-Kamiokande measurement (Abe et al., 2016).	36
13	^8B survival probability $P_{ee}(E)$ under the Standard Model and several representative NSI presets (left), and the resulting distortion relative to the Standard Model, in percentage points (right).	37
14	Predicted (tpeanuts , at Daya Bay’s own published truth) vs. real observed prompt-energy spectra, near-hall (AD11) and far-hall (AD31) detectors.	39

List of Tables

1	Software layers of tpeanuts and their responsibility.	19
2	Experiments implemented on the detector/inference layers, and their validation status against real published data (section 5.7).	21
3	Benchmark observables reproduced from the atmospheric oscillogram, compared with published experimental results.	35
4	Confirmed 3+1 sterile-extension validation results (notebooks/validation/physics/BSM/sterile1_t	
5	Daya Bay closure test: predicted vs. observed IBD counts at Daya Bay’s own published $(\theta_{13}, \Delta m_{31}^2)$, no fit.	38
6	Daya Bay near/far ratio fit vs. Daya Bay’s own published best fit and NuFIT 6.1.	39
7	Notebook directory tree (paths relative to notebooks/ in the repository above). .	44

1 Introduction

Neutrino oscillations are one of the clearest experimental indications that the Standard Model of particle physics must be extended, as established by the atmospheric-neutrino measurements of Super-Kamiokande and the solar-neutrino measurements of the Sudbury Neutrino Observatory (Ahmad et al., 2002; Fukuda et al., 1998). A neutrino produced with a definite flavour, such as an electron, muon or tau neutrino, propagates as a coherent superposition of mass eigenstates. The interference among these components gives rise to flavour transitions whose probabilities depend on the neutrino energy, the travelled distance, the mixing parameters and, when propagation occurs in matter, the electron density of the medium.

Modern neutrino analyses require evaluating these probabilities many times. Oscillation probabilities appear inside flux predictions, detector simulations, likelihood scans and global fits. For this reason, the numerical implementation of the oscillation formalism is not a secondary detail: it directly affects the feasibility of high-statistics studies and of exploratory scans over large parameter spaces. Existing tools such as PEANUTS (Gonzalo & Lucente, 2024) provide compact routines for neutrino propagation, matter mixing and solar-neutrino probabilities. However, array-based implementations are not always optimal for current workflows involving GPUs, TPUs, automatic differentiation and batched execution.

This thesis studies and documents the development of `tpeanuts`, a PEANUTS update oriented towards PyTorch (Paszke et al., 2019). The goal is not to change the physical model implemented by PEANUTS, but to preserve its conventions while expressing its numerical core in a tensor framework, and to build on top of it a detector-level forward-folding and statistical-inference layer capable of fitting real experimental data. This makes it possible to evaluate many oscillation configurations in parallel, to move computations between CPU and GPU with minimal changes, and to connect the resulting probabilities to gradient-based parameter estimation. Beyond this core translation, `tpeanuts` pursues a modular design that extends the original PEANUTS implementation with support for atmospheric neutrinos, new-physics scenarios, and a chain of detector and inference models validated against several real experiments.

1.1 Motivation

The main motivation for `tpeanuts` is the need for fast, repeated neutrino-oscillation evaluation, both for large-scale simulation (atmospheric-flux studies, detector-response studies, BSM scans) and, ultimately, for fitting real data: a gradient-based fit to a detector’s published spectrum requires the oscillation probability, the detector response and the likelihood to be evaluated, and differentiated, at every optimiser step. PyTorch provides a natural environment for this because it combines vectorised tensor operations, GPU support and automatic differentiation, letting a single implementation serve fast forward simulation and gradient-based inference alike.

The original PEANUTS implementation is tied to a specific solar model and a fixed treatment of matter effects, which limits its use where the sensitivity of a result to these choices must be tested. Building `tpeanuts` on a flexible tensor framework makes it possible to add such options, including alternative solar and Earth density models, perturbative treatments of the matter potential, and new-physics scenarios such as non-standard interactions (NSI) and sterile neutrinos, in a modular

way, without compromising the validated core of the oscillation formalism. This same modularity is what makes it practical to add a detector-response layer (target composition, energy resolution, backgrounds, efficiencies) and an inference layer (likelihood construction, gradient-based fitting, uncertainty estimation) on top of the oscillation engine, and to reuse both across experiments as different as a reactor (Daya Bay), a liquid-scintillator solar detector (Borexino), a large-volume atmospheric detector (IceCube DeepCore) and a heavy-water Cherenkov detector (SNO).

A further, complementary motivation comes from atmospheric-neutrino analyses, where realistic differential flux structures across energy and zenith angle are costly to reproduce from full cascade simulations; this motivates a diffusion-based generative model, developed alongside **tpeanuts**, as a fast, learned alternative within the same PyTorch ecosystem.

1.2 Objectives

The general objective of this Master's Thesis is to develop and document **tpeanuts**, a PyTorch update of PEANUTS for fast neutrino simulation, and its extension into a detector-level statistical-inference framework validated against real experimental data. The specific objectives are:

- to review the three-flavour oscillation formalism used by PEANUTS, including vacuum propagation and matter effects, and to translate its numerical core to PyTorch tensors while preserving physical conventions, units and sign choices;
- to design a modular architecture in which solar/Earth models, matter and perturbative treatments, and new-physics scenarios (NSI, sterile neutrinos) can be selected and reused consistently across the application;
- to define a validation strategy comparing **tpeanuts** against PEANUTS and **nuSQuIDS** at the level of intermediate quantities and final probabilities, complemented by the reproduction of well-known theoretical and experimental results and by a CPU/GPU benchmarking methodology;
- to implement a detector-agnostic forward-folding layer (cross sections, target composition, energy response, efficiency, backgrounds) and to specialise it for real experiments;
- to implement a gradient-based statistical-inference layer (Poisson and Gaussian likelihoods, LBFGS fitting, Laplace/Fisher uncertainty estimation, likelihood scans) built on PyTorch autograd;
- to validate the full detector-inference chain against real, published data, with Daya Bay as the primary case study, and to document its application to Borexino, IceCube DeepCore and SNO;
- to design and train a diffusion-based generative model synthesising differential atmospheric-neutrino flux structures, as a complementary fast-simulation component.

1.3 Document structure

Section 2 introduces the theoretical framework of neutrino masses, mixing, oscillations, matter effects and the statistical-inference formalism. Section 3 describes the methodology used to translate, validate and fit the numerical implementation. Section 4 presents the technical development of `tpeanuts`, including its detector and inference layers. Section 5 presents validation outputs, benchmark results, and the reproduction of physical results, closing with the Daya Bay detector-level case study. Finally, Section 6 summarises the conclusions and outlines future work.

2 Theoretical Framework

This section summarises the physical ingredients required by `tpeanuts`. The objective is not to provide a complete review of neutrino physics, but to establish the conventions that must be preserved when updating PEANUTS to a tensor-based implementation.

2.1 Neutrino propagation fundamentals

This subsection collects the formalism common to every propagation regime considered in this thesis: the definition of the mixing matrix, vacuum oscillations, and the modification of the propagation Hamiltonian by ordinary matter. The specific neutrino sources and new-physics extensions built on top of this common core are treated separately in the following subsections.

2.1.1 Neutrino mixing

One of the fundamental discoveries in particle physics is that neutrinos are produced and detected in flavour eigenstates, whereas they propagate through space as mass eigenstates. Since these two bases do not coincide, a neutrino created with a definite flavour evolves into a quantum superposition of different flavours during propagation, giving rise to the phenomenon of neutrino oscillations (Giunti & Kim, 2007).

Within the Standard Model extended to include neutrino masses, the flavour eigenstates $|\nu_\alpha\rangle$, $\alpha = e, \mu, \tau$, are related to the mass eigenstates $|\nu_i\rangle$, $i = 1, 2, 3$, through a unitary transformation known as the Pontecorvo–Maki–Nakagawa–Sakata (PMNS) matrix (Maki et al., 1962; Pontecorvo, 1957, 1958),

$$|\nu_\alpha\rangle = \sum_{i=1}^3 U_{\alpha i}^* |\nu_i\rangle, \quad (1)$$

or, equivalently,

$$|\nu_i\rangle = \sum_{\alpha=e,\mu,\tau} U_{\alpha i} |\nu_\alpha\rangle. \quad (2)$$

The matrix U is unitary, $U^\dagger U = U U^\dagger = \mathbb{I}$, which guarantees the conservation of probability during neutrino propagation.

Physical interpretation. The matrix elements $U_{\alpha i}$ represent the probability amplitudes that a neutrino with flavour α is found in the mass eigenstate i . Consequently, $|U_{\alpha i}|^2$ gives the probability that a neutrino produced as flavour α propagates in the mass eigenstate i . This mismatch between flavour and mass eigenstates is directly analogous to the quark sector, where flavour mixing is described by the Cabibbo–Kobayashi–Maskawa (CKM) matrix (Workman et al., 2022). In the lepton sector, however, the mixing angles are significantly larger, making flavour oscillations observable over laboratory, atmospheric, solar and astrophysical baselines.

Standard parametrisation. For three active Dirac neutrinos, the PMNS matrix is written as the product of three elementary rotations, $U_{\text{PMNS}} = R_{23}(\theta_{23}) U_{13}(\theta_{13}, \delta_{\text{CP}}) R_{12}(\theta_{12})$, yielding the standard Particle Data Group (PDG) parametrisation (Workman et al., 2022)

$$U_{\text{PMNS}} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta_{\text{CP}}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta_{\text{CP}}} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta_{\text{CP}}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta_{\text{CP}}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta_{\text{CP}}} & c_{23}c_{13} \end{pmatrix}, \quad (3)$$

with $c_{ij} = \cos \theta_{ij}$, $s_{ij} = \sin \theta_{ij}$. Each rotation has a clear physical meaning: θ_{12} governs mixing between the first and second mass eigenstates and dominates solar-neutrino oscillations together with the smaller mass splitting Δm_{21}^2 ; θ_{23} controls mixing between the second and third eigenstates and is primarily responsible for atmospheric-neutrino oscillations; θ_{13} couples the first and third eigenstates and, although considerably smaller than the other mixing angles, plays a crucial role in matter effects and enables sensitivity to CP violation. The complex phase δ_{CP} introduces a difference between neutrino and antineutrino oscillation probabilities, providing a possible source of CP violation in the lepton sector.

Oscillation parameters. Oscillation experiments are sensitive not to the absolute neutrino masses but to the differences between their squared masses, $\Delta m_{ij}^2 = m_i^2 - m_j^2$. Only two independent mass-squared differences exist, Δm_{21}^2 and Δm_{31}^2 (or equivalently Δm_{32}^2), together with the three mixing angles and the CP-violating phase,

$$\{\theta_{12}, \theta_{13}, \theta_{23}, \Delta m_{21}^2, \Delta m_{31}^2, \delta_{\text{CP}}\}. \quad (4)$$

These six quantities completely determine neutrino oscillation probabilities within the standard three-neutrino oscillation framework. Current best-fit values are obtained through global analyses that combine measurements from solar, atmospheric, reactor and accelerator experiments. The most widely used compilation is provided by the NuFIT collaboration (Esteban et al., 2022), whose parameters constitute the reference input for the calculations presented throughout this work.

2.1.2 Vacuum oscillations

Once produced in a flavour eigenstate, neutrinos propagate as coherent superpositions of mass eigenstates. Since each mass eigenstate accumulates a different quantum phase, their interference is the physical origin of neutrino oscillations (Giunti & Kim, 2007). The evolution is governed by

the Schrödinger equation,

$$i \frac{d}{dt} |\nu(t)\rangle = H |\nu(t)\rangle, \quad (5)$$

where H is the effective Hamiltonian describing neutrino propagation.

Vacuum Hamiltonian. In vacuum, H is diagonal in the mass basis, $H_{\text{mass}} = \text{diag}(E_1, E_2, E_3)$, with $E_i = \sqrt{p^2 + m_i^2}$. Since neutrinos produced in laboratory, solar and atmospheric experiments are highly relativistic, $m_i^2 \ll p^2$, the energy can be expanded as $E_i \simeq p + m_i^2/(2E)$, with $E \simeq p$ the neutrino energy. The common momentum term contributes only an overall global phase, e^{-ipL} , which has no observable consequences and can be dropped, leaving the effective vacuum Hamiltonian

$$H_{\text{vac}} = \frac{1}{2E} \text{diag}(m_1^2, m_2^2, m_3^2). \quad (6)$$

Flavour evolution requires transforming H_{vac} into the flavour basis through the PMNS matrix,

$$H_f = U H_{\text{vac}} U^\dagger = \frac{1}{2E} U \text{diag}(m_i^2) U^\dagger. \quad (7)$$

Evolution operator. For a constant Hamiltonian, the Schrödinger equation is solved by $|\nu(L)\rangle = e^{-iHL} |\nu(0)\rangle$. For a neutrino produced in flavour α , $|\nu(0)\rangle = |\nu_\alpha\rangle$, the transition amplitude to flavour β after a distance L is

$$A_{\alpha \rightarrow \beta}(L, E) = \sum_{i=1}^3 U_{\beta i} \exp\left(-i \frac{m_i^2 L}{2E}\right) U_{\alpha i}^*, \quad (8)$$

with oscillation probability $P_{\alpha \rightarrow \beta} = |A_{\alpha \rightarrow \beta}|^2$. Unitarity of the PMNS matrix guarantees $\sum_\beta P_{\alpha \rightarrow \beta} = 1$, i.e. conservation of probability. PEANUTS implements this evolution with dimensionless distances and phases, using the Earth radius as a convenient reference scale in some routines.

Oscillation phases. Oscillations arise from the phase differences accumulated by the mass eigenstates, controlled by the mass-squared differences $\Delta m_{ij}^2 = m_i^2 - m_j^2$, which define the oscillation phase $\Delta_{ij} = \Delta m_{ij}^2 L / 2E$, or, in the practical units customary in oscillation phenomenology (Workman et al., 2022), $\Delta_{ij} = 1.267 \Delta m_{ij}^2 [\text{eV}^2] L [\text{km}] / E [\text{GeV}]$; oscillations are thus governed by the ratio L/E rather than by the baseline or the energy separately.

2.1.3 Matter effects

The vacuum formalism above assumes that neutrinos propagate through empty space. In many settings relevant to `tpeanuts`, such as the solar interior, the Earth's mantle and core, or a supernova envelope, neutrinos instead traverse dense media, where coherent forward scattering off the particles of the medium modifies the propagation Hamiltonian. This is the matter effect, also known as the Mikheyev–Smirnov–Wolfenstein (MSW) effect (Giunti & Kim, 2007; Mikheyev

& Smirnov, 1985; Wolfenstein, 1978). Because coherent forward scattering transfers essentially no momentum to the medium, it does not remove neutrinos from the beam; instead, it modifies the phase accumulated during propagation, which is described as an additional effective potential in the Schrödinger equation.

Origin of the matter potential. Neutrinos interact with matter through both neutral-current (NC) and charged-current (CC) weak processes. NC interactions, $\nu_\alpha + f \rightarrow \nu_\alpha + f$ with f an electron, proton or neutron, affect all active flavours equally; since oscillation probabilities depend only on relative phases between flavours, this common contribution is unobservable and is omitted from the oscillation Hamiltonian. Electron neutrinos additionally undergo the CC process $\nu_e + e^- \rightarrow \nu_e + e^-$, mediated by $W^\pm$ exchange, which introduces a flavour-dependent potential

$$V_{\text{CC}} = \sqrt{2} G_F n_e, \quad (9)$$

where G_F is the Fermi coupling constant and n_e the local electron number density. This additional, electron-flavour-specific interaction breaks the flavour symmetry of vacuum propagation and is the origin of the MSW effect.

Effective Hamiltonian. The full Hamiltonian governing flavour evolution in matter is obtained by adding the matter potential to the vacuum Hamiltonian,

$$H = H_{\text{vac}} + V_{\text{mat}}, \quad H_{\text{vac}} = \frac{1}{2E} U \text{diag}(m_1^2, m_2^2, m_3^2) U^\dagger, \quad V_{\text{mat}} = \sqrt{2} G_F n_e \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (10)$$

where V_{mat} is diagonal in the flavour basis, since only the electron flavour experiences the CC interaction; for antineutrinos it changes sign, $V_{\text{mat}} \rightarrow -V_{\text{mat}}$ (section 4.2.2). In matter, the propagation basis becomes density-dependent, $H = U_m \text{diag}(\lambda_1, \lambda_2, \lambda_3) U_m^\dagger$, with U_m the effective mixing matrix and λ_i the effective eigenvalues; as the electron density varies along the trajectory, both evolve continuously, responsible for the characteristic energy dependence of solar-neutrino oscillations.

2.2 Neutrino sources

The oscillation formalism introduced above is universal: the same Schrödinger equation governs flavour evolution regardless of how the neutrino is produced. What varies between sources is the physical environment of production and, consequently, the density profile traversed before detection, which determines the relative importance of vacuum oscillations, matter effects and decoherence. Each source is therefore sensitive to a different region of the oscillation parameter space.

The current version of `tpeanuts` is designed primarily for the simulation of solar and atmo-

spheric neutrinos, which together probe a broad range of energies, baselines and matter densities. Owing to its modular architecture, however, the propagation engine itself is independent of the neutrino source, so that additional sources can be incorporated with minimal modifications, as discussed for reactor and accelerator neutrinos below.

2.2.1 Solar neutrinos

Solar neutrinos are produced by thermonuclear fusion in the solar core, where temperatures reach $\sim 1.5 \times 10^7$ K and densities exceed 150 g cm^{-3} . The dominant production mechanisms are the proton–proton (pp) chain and the CNO cycle, generating neutrinos from a few hundred keV up to ~ 20 MeV.

Unlike reactor or accelerator neutrinos, solar neutrinos are produced deep inside a medium whose density decreases continuously from the core to the photosphere, so that flavour evolution is strongly affected by the MSW effect. For the Large Mixing Angle (LMA) solution favoured by current oscillation data, propagation through the Sun is highly adiabatic over most of the relevant energy range: the neutrino remains in the same instantaneous matter eigenstate as the local density changes (Haxton, 1986). Within this approximation, the flavour state produced at radius r is projected onto the local matter eigenstates, which evolve adiabatically into vacuum mass eigenstates as the neutrino exits the Sun. After propagating over the Earth–Sun distance (1 AU), the mass eigenstates lose mutual coherence due to wave-packet separation and the finite energy resolution of detectors, so the flavour transition probabilities reduce to an incoherent sum,

$$P_{\nu_e \rightarrow \nu_\alpha}^{(f)}(E) = \sum_{i=1}^3 |U_{\alpha i}|^2 T_i^{(f)}(E), \quad (11)$$

where $T_i^{(f)}(E)$ is the probability that a neutrino produced through solar reaction f emerges from the Sun as vacuum mass eigenstate ν_i . Since production is spatially distributed throughout the solar interior, $T_i^{(f)}$ is obtained by averaging over the corresponding radial production profile,

$$T_i^{(f)}(E) = \int_0^{R_\odot} w_f(r) P_{\nu_e \rightarrow \nu_i}(E, r) dr, \quad (12)$$

with $w_f(r)$ the normalised production distribution predicted by the Standard Solar Model (SSM). The solar module in **tpeanuts** uses radial electron-density profiles together with production distributions for the principal channels (pp , pep , hep , ${}^7\text{Be}$, ${}^8\text{B}$, ${}^{13}\text{N}$, ${}^{15}\text{O}$, ${}^{17}\text{F}$), using standard solar models such as B16(GS98) and B16(AGSS09met) (Vinyoles et al., 2017). Optional non-adiabatic corrections based on the Landau–Zener formalism (Parke, 1986) are also available for the regions where the adiabatic approximation becomes less accurate. Because solar neutrinos experience both matter-enhanced conversion inside the Sun and, at night, regeneration inside the Earth, they are one of the most powerful probes of neutrino propagation in matter and provide stringent constraints on the solar mixing parameters and on possible non-standard interactions.

2.2.2 Atmospheric neutrinos

Atmospheric neutrinos are produced when high-energy cosmic rays interact with nuclei in the atmosphere, generating hadronic cascades dominated by pions and kaons; the decay of these mesons and of the resulting secondary muons produces both muon and electron neutrinos, spanning energies from a few hundred MeV to beyond the TeV scale (Honda et al., 2007). Unlike solar neutrinos, they are produced over a wide range of altitudes and arrive at terrestrial detectors from all directions, with baselines extending from ~ 20 km for downward-going events to nearly the Earth’s diameter ($\sim 12\,742$ km) for upward-going neutrinos that cross the planet.

This broad coverage in energy and baseline makes atmospheric neutrinos sensitive to several oscillation regimes at once: downward-going events are dominated by vacuum oscillations, while upward-going neutrinos experience significant matter effects while traversing the Earth’s mantle and core. The interplay between the two regimes produces the characteristic oscillation patterns in the $(E, \cos \theta_z)$ plane known as oscillograms. Within **tpeanuts**, atmospheric-neutrino calculations combine the three-flavour propagation formalism with realistic Earth density profiles, such as the Preliminary Reference Earth Model (PREM) (Dziewonski & Anderson, 1981), and differential atmospheric flux tables from the Honda parametrisation (Honda et al., 2007) or from cascade solvers such as MCEq (Fedynitch et al., 2015). Beyond serving as physical inputs to the oscillation calculation, these flux structures are also the target distributions for the diffusion-based generative model developed alongside **tpeanuts**, which aims to produce fast, high-fidelity approximations of computationally expensive atmospheric cascade simulations.

2.2.3 Reactor antineutrinos

Nuclear reactors are intense, well-characterised sources of electron antineutrinos, emitted in the beta decays of the neutron-rich fission fragments of ^{235}U , ^{238}U , ^{239}Pu and ^{241}Pu , with a spectrum extending up to ~ 8 MeV and a rate set by the reactor’s thermal power. Because flavour evolution is entirely determined by the Hamiltonian of section 2.1, and reactor baselines ($\lesssim 2$ km for a near/far experiment such as Daya Bay) traverse negligible Earth matter, reactor antineutrinos are described by the exact coherent three-flavour *vacuum* survival probability of section 2.1, with θ_{13} and Δm_{31}^2 as the parameters to which the disappearance channel $\bar{\nu}_e \rightarrow \bar{\nu}_e$ is sensitive. **tpeanuts** implements a full reactor-antineutrino observable chain on top of this probability: the per-isotope Huber–Mueller emission spectrum (Huber, 2011; Mueller et al., 2011), including non-equilibrium and spent-nuclear-fuel corrections to the fission fractions, the inverse-beta-decay (IBD) cross section to order $1/M$ (Vogel & Beacom, 1999), and a detector-response chain (energy-scale non-linearity, light-yield non-uniformity and Gaussian energy resolution) folding the true antineutrino spectrum into an observable prompt-energy spectrum. This chain, together with the corresponding detector layer, is described in section 4.1.6 and validated against real Daya Bay data in section 5.6.

2.2.4 Other neutrino sources

Accelerator-based experiments produce controlled muon-neutrino or antineutrino beams over baselines from a few hundred to more than a thousand kilometres, enabling precision measurements

of the atmospheric oscillation parameters, the CP-violating phase and the neutrino mass ordering. As with reactor antineutrinos, incorporating such a source requires only specifying the appropriate initial flavour composition, energy spectrum, baseline and matter-density profile within the same propagation engine, without any change to the oscillation formalism itself.

2.3 New physics

Beyond-the-Standard-Model scenarios can be incorporated into the same formalism through modifications of the effective Hamiltonian and, in the case of sterile neutrinos, of the PMNS matrix itself. Two such extensions are considered in **tpeanuts**: non-standard interactions and sterile neutrinos.

2.3.1 Non-standard interactions (NSI)

Although the three-flavour framework introduced above explains the vast majority of neutrino oscillation observations, it is widely regarded as an effective low-energy description rather than a complete theory: the Standard Model does not account for the origin of neutrino masses, nor does it explain the nature of dark matter, the matter–antimatter asymmetry or the hierarchy of fermion masses. Many theories beyond the Standard Model (BSM) therefore predict additional neutrino interactions that would manifest as small deviations from the standard oscillation paradigm.

Among the most studied such scenarios are non-standard interactions (NSI), in which neutrinos experience additional weak-scale interactions with matter mediated by new heavy particles (Grossman, 1995). At energies well below the mediator mass scale, these effects can be described with an effective field theory in which the heavy degrees of freedom are integrated out and encoded in dimensionless effective couplings, providing a model-independent framework for a broad class of BSM scenarios without specifying the underlying microscopic theory.

NSI can in general affect neutrino production, detection and propagation; this work considers only *propagation* NSI, in which new neutral-current-like interactions modify the coherent forward scattering of neutrinos through matter. Since coherent forward scattering only changes the propagation phase, these interactions appear as an additional contribution to the matter Hamiltonian,

$$H_{\text{mat}} = \sqrt{2}G_F n_e \left[\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} \varepsilon_{ee} & \varepsilon_{e\mu} & \varepsilon_{e\tau} \\ \varepsilon_{e\mu}^* & \varepsilon_{\mu\mu} & \varepsilon_{\mu\tau} \\ \varepsilon_{e\tau}^* & \varepsilon_{\mu\tau}^* & \varepsilon_{\tau\tau} \end{pmatrix} \right], \quad (13)$$

where the dimensionless couplings $\varepsilon_{\alpha\beta}$ quantify the strength of the non-standard interaction relative to the Standard Model weak interaction. The diagonal terms rescale the effective matter potential felt by each flavour, while the off-diagonal terms introduce flavour-changing coherent interactions with no Standard Model analogue. The complete propagation Hamiltonian is then $H = H_{\text{vac}} + H_{\text{mat}}$, with $H_{\text{vac}} = \frac{1}{2E}UM^2U^\dagger$ retaining the standard PMNS structure unchanged. NSI therefore leave vacuum oscillations unaffected but modify flavour evolution whenever neutrinos propagate through matter, with an impact that depends on both the local

density and the neutrino energy, making long-baseline, atmospheric and solar experiments particularly sensitive probes.

Phenomenologically, NSI modify the effective mixing angles and mass splittings in matter, shift the MSW resonance, alter the day–night regeneration of solar neutrinos, and can introduce degeneracies with the standard oscillation parameters. A well-known example is the LMA-Dark solution, where a large negative ε_{ee} compensates a transformation of the solar mixing angle, yielding oscillation probabilities nearly indistinguishable from the standard Large Mixing Angle (LMA) solution; such degeneracies illustrate the value of combining different classes of neutrino experiments to disentangle standard oscillation physics from possible BSM effects. Current data constrain any NSI to be subdominant with respect to the Standard Model weak interaction, although several couplings, particularly those involving the τ flavour, remain only weakly bounded; global analyses combining oscillation, scattering and collider measurements provide increasingly stringent limits on the allowed parameter space (Biggio et al., 2009; Esteban et al., 2018).

From the computational point of view, NSI are a natural extension of the propagation formalism implemented in **tpeanuts**: since they modify only the matter contribution to the Hamiltonian, the numerical propagation algorithms developed for the standard three-flavour model remain unchanged. The implementation introduces a fully general complex Hermitian NSI matrix, allowing both flavour-conserving and flavour-changing couplings, which enables direct comparison between Standard Model predictions and a broad range of BSM scenarios while preserving the efficiency and flexibility of the underlying PyTorch implementation.

2.3.2 Sterile neutrinos

The Standard Model contains three active flavours, ν_e , ν_μ and ν_τ , all of which participate in the weak interaction through charged- and neutral-current processes. Several BSM models, together with a number of experimental anomalies reported over the past decades, have nonetheless motivated the existence of additional neutrino states that are singlets under the Standard Model gauge group. Because they do not couple directly to the weak interaction, these hypothetical states are known as *sterile* neutrinos (Böser et al., 2020). Although they cannot be produced or detected through Standard Model weak processes, sterile neutrinos may mix with the active flavours through the neutrino mass matrix and thus participate in flavour oscillations as additional propagation eigenstates, modifying the observed oscillation pattern without introducing any new interaction at the detector.

The simplest and most widely studied extension is the $3 + 1$ model, in which a single sterile state is added to the three active flavours. The flavour basis becomes $(\nu_e, \nu_\mu, \nu_\tau, \nu_s)$ and the propagation basis four mass eigenstates $(\nu_1, \nu_2, \nu_3, \nu_4)$, so that the 3×3 PMNS matrix is enlarged into a 4×4 unitary mixing matrix, $U_4 = \text{diag}(U_{\text{PMNS}}, 1) R_{34}(\theta_{34}) R_{24}(\theta_{24}) R_{14}(\theta_{14})$, where the rotations R_{i4} introduce mixing between the sterile state and each active flavour. Beyond the standard three-flavour parameters, the $3 + 1$ framework thus introduces three new mixing angles $\theta_{14}, \theta_{24}, \theta_{34}$, a fourth mass eigenstate m_4 , and a new mass-squared difference $\Delta m_{41}^2 = m_4^2 - m_1^2$, typically assumed $\gg |\Delta m_{31}^2|$ (of order 1 eV^2) in phenomenological analyses motivated by short-baseline anomalies, producing oscillation lengths much shorter than the atmospheric or solar

ones.

Unlike NSI, which modify only the matter contribution to the propagation Hamiltonian, sterile neutrinos change the formalism at a more fundamental level: the vacuum Hamiltonian itself must be enlarged,

$$H_{\text{vac}} = \frac{1}{2E} U_4 \text{diag}(m_1^2, m_2^2, m_3^2, m_4^2) U_4^\dagger, \quad (14)$$

so that the new eigenstate contributes its own oscillation phase, $\exp(-im_4^2 L/2E)$, to the evolution operator. Matter effects are modified as well: since the sterile state does not participate in Standard Model weak interactions, it experiences neither the charged-current potential nor the neutral-current (NC) potential common to the three active flavours, $V_{\text{NC}} = \mp \frac{1}{\sqrt{2}} G_F n_n L_{\text{scale}}$ (from the neutron density n_n), which is an unobservable global phase in the pure three-flavour case but becomes a genuine, observable relative potential once this NC-blind sterile state is added: after subtracting the common phase,

$$H_{\text{mat}} = \text{diag}(V_{\text{CC}}, 0, 0, -V_{\text{NC}}), \quad (15)$$

with $V_{\text{CC}} = \sqrt{2} G_F n_e L_{\text{scale}}$. **tpeanuts** implements this NC term as an optional addition to the sterile Hamiltonian (`core/common/potential.py::matter_potential_nc`), active whenever a neutron-density profile is supplied; omitting it recovers the CC-only sterile matter term used throughout most of the sterile-oscillation literature, an approximation rather than an exact simplification once θ_{24} or θ_{34} are non-zero. Either way, the modified matter Hamiltonian alters the effective propagation eigenvalues and can generate additional resonant conversion phenomena in dense media.

Phenomenologically, sterile neutrinos may produce anomalous flavour disappearance, unexpected appearance channels, distortions of oscillation spectra and modified matter-induced effects, and have been extensively investigated in connection with the LSND and MiniBooNE anomalies, the reactor antineutrino anomaly and the Gallium anomaly, although no conclusive evidence has yet established their existence (Böser et al., 2020). Cosmological observations independently constrain light sterile neutrinos through their impact on Big Bang nucleosynthesis, the effective number of relativistic degrees of freedom, and large-scale structure formation.

Within **tpeanuts**, sterile neutrinos are implemented as a natural extension of the standard three-flavour propagation framework: the oscillation engine has been generalised to support arbitrary Hamiltonian dimensions, so that the 4×4 mixing matrix, the enlarged mass spectrum and the corresponding matter Hamiltonian are treated with the same numerical propagation algorithms developed for the standard model. The current implementation fully supports the $3 + 1$ scenario, while its modular design allows straightforward extension to models with multiple sterile states or more general BSM oscillation frameworks.

2.4 Statistical inference formalism

Comparing an oscillation prediction against real, binned experimental data requires a statistical framework connecting predicted and observed counts, a rule for minimising the resulting statistic

with respect to the oscillation and detector parameters, and a way of estimating the uncertainty on the fitted values. **tpeanuts** implements this framework directly on top of PyTorch’s automatic differentiation, so that the same gradient used to build the oscillation Hamiltonian also drives the fit.

For a set of bins with predicted counts $N_k(\theta)$ and observed counts n_k^{obs} , the Poisson (Baker–Cousins) likelihood-ratio statistic (Baker & Cousins, 1984)

$$-2 \ln L_{\text{Poisson}} = 2 \sum_k \left[N_k(\theta) - n_k^{\text{obs}} + n_k^{\text{obs}} \ln \frac{n_k^{\text{obs}}}{N_k(\theta)} \right] \quad (16)$$

is used for absolute event counts, and vanishes exactly when $N_k = n_k^{\text{obs}}$ bin by bin. For derived observables that are not themselves Poisson-distributed, such as a ratio of counts, a Gaussian statistic with propagated uncertainty is used instead, $-2 \ln L_{\chi^2} = \sum_k [(R_k(\theta) - R_k^{\text{obs}})/\sigma_k]^2$. Externally measured nuisance parameters (background normalisations, energy-scale pulls) are constrained by an additional Gaussian penalty term on θ itself, added to either statistic.

The best-fit point $\hat{\theta} = \arg \min_{\theta} [-2 \ln L(\theta)]$ is found with the quasi-Newton **LBFGS** optimiser, using the exact gradient of the total statistic obtained by backpropagating through the full physics chain, oscillation probability, detector response and likelihood, rather than finite differences. The uncertainty on $\hat{\theta}$ is estimated through the Laplace approximation, in which the Fisher information at the optimum, $\mathcal{I} = \frac{1}{2} \nabla_{\theta}^2 (-2 \ln L)$, obtained as the autograd Hessian of the statistic, gives the parameter covariance $\Sigma = \mathcal{I}^{-1}$, with $\sigma_j = \sqrt{\Sigma_{jj}}$. Where the local Laplace approximation is inadequate, in particular along directions the data barely constrain, a direct two-dimensional scan of $-2 \ln L$ over a parameter grid, optionally profiling the remaining free parameters at each grid point, provides confidence contours calibrated through Wilks’ theorem.

2.5 Diffusion generative model

As noted above, the differential atmospheric-neutrino flux, resolved in energy, zenith angle and atmospheric production height, is costly to obtain from full cascade simulations, motivating a fast, learned surrogate. **tpeanuts** explores this surrogate as a denoising diffusion probabilistic model (DDPM) (Ho et al., 2020; Sohl-Dickstein et al., 2015), a generative model that learns to reverse a fixed, gradual noising process rather than sampling the target distribution directly.

The forward process is a Markov chain that corrupts a clean flux tensor x_0 into Gaussian noise over T steps according to a variance schedule $\{\beta_t\}_{t=1}^T$. Because Gaussians compose, the marginal at any step t follows in closed form, with $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$,

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbb{I}), \quad (17)$$

so that, for a sufficiently long schedule, x_T is indistinguishable from pure noise. Generation reverses this chain, starting from Gaussian noise and iteratively denoising; Ho et al. (2020) show that, under a suitable reparametrisation of the variational bound, the reverse transitions reduce to training a network $\epsilon_{\theta}(x_t, t)$ to predict the noise injected at each step,

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{x_0, \epsilon, t} \left[\left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2 \right], \quad (18)$$

with equivalent reparametrisations targeting x_0 or the noise/signal velocity offering better numerical stability at extreme noise levels, and a cosine variance schedule commonly preferred over the original linear one to avoid concentrating destructive noise at the earliest steps (Nichol & Dhariwal, 2021).

Because the target flux depends on the physical configuration of each shower rather than being drawn from an unconditional population, the model is trained as a *conditional* generative model, $p_\theta(x_0 | c)$, with an auxiliary vector c of simulation parameters supplied to the denoising network alongside the noisy sample and the timestep. The network itself follows the convolutional encoder–decoder U-Net architecture originally introduced for image segmentation (Ronneberger et al., 2015), generalised to the three-dimensional flux-tensor grid, with the timestep injected through sinusoidal embeddings analogous to positional encodings. At inference time, samples are drawn either with the ancestral DDPM sampler, which evaluates the network once per diffusion step, or with the deterministic, non-Markovian sampler of Song et al. (2021) (DDIM), which reaches compatible marginals from a small subset of timesteps at a fraction of the cost.

2.6 Related computational tools

tpeanuts interfaces with several public tools, either for cross-validation or as external inputs. **nuSQuIDS** (Arguelles et al., 2022), a general framework for solving neutrino quantum kinetic equations, is used as an independent, actively maintained numerical solver against which vacuum, solar and atmospheric probabilities are cross-validated; **PEANUTS** remains lighter and more specialised, precisely the property **tpeanuts** aims to preserve. **MCEq** (Fedynitch et al., 2015), which solves the coupled cascade equations of cosmic-ray-induced air showers, and the **Honda/HKKM** tables (Honda et al., 2007) provide the two atmospheric-neutrino flux inputs, **MCEq** additionally supplying training data for the diffusion-based generative model of section 2.5; the companion **CRFlux** library supplies the primary cosmic-ray flux parametrisations (e.g. Polygonato) shared by **MCEq** and the **CORSIKA**-based sampling pipeline, so that both remain anchored to the same top-of-atmosphere spectrum. **pymis** (Emmert et al., 2021) provides empirical atmospheric density/composition profiles as a function of altitude, location and time, replacing constant-density approximations. **CORSIKA** (Heck et al., 1998) is used in an exploratory capacity to generate primary cosmic-ray samples and shower-level neutrino yields.

3 Methodology

The methodology followed in this work combines physics validation and software translation. Since **tpeanuts** is intended as an update of **PEANUTS**, correctness is defined primarily by agreement with the original implementation under the same conventions, parameters and numerical tolerances.

3.1 Translation methodology

The development of `tpeanuts` follows a systematic translation process in which the original PEANUTS implementation is progressively transformed into a fully tensor-based PyTorch framework while preserving its physical predictions. Rather than a direct line-by-line conversion, the methodology separates the physical algorithms from implementation-specific details, so the resulting architecture remains both computationally efficient and scientifically transparent; numerical validation against the reference implementation is treated separately in section 3.2, and the resulting modular software architecture in section 4.1.6.

3.1.1 Analysis and translation

Each PEANUTS module (PMNS construction, vacuum and matter Hamiltonians, matter mixing, numerical propagation, solar production and Earth propagation, detector exposure) is first interpreted from the underlying theoretical formalism of section 2.1, separating implementation-specific assumptions, array layouts, scalar-oriented routines, interpolation methods, from the physical algorithm itself. Each procedure is then reformulated using native, batched PyTorch tensor operations, so that arbitrarily large grids of energies, baselines and production points are evaluated within a single vectorised call: one-dimensional integrations use `torch.trapezoid`, tabulated profiles are interpolated differentiably, and numerical constants are stored as configurable-precision, configurable-device tensors, avoiding NumPy–PyTorch conversions that would otherwise force CPU–GPU data transfers.

3.1.2 Verification of units and physical conventions

Maintaining consistency with PEANUTS’s physical conventions, the PMNS phase convention, the neutrino/antineutrino sign, mass-eigenstate ordering, matter-potential normalisation, and the dimensionless propagation variables, is a critical, explicitly verified part of the translation, since sign errors or unit-conversion mistakes can produce oscillation probabilities that remain numerically stable while being physically incorrect. For every translated module, intermediate quantities (mixing matrices, Hamiltonians, eigenvalues, matter potentials, oscillation phases) are compared directly against the original implementation before validating the final oscillation probabilities.

3.2 Validation strategy

Validation begins with numerical, code-to-code agreement against the reference PEANUTS and `nuSQuIDS` implementations. This is complemented by the reproduction of well-known experimental and theoretical results in neutrino physics, described separately in section 3.3.

3.2.1 Comparison with PEANUTS and nuSQuIDS

The primary validation strategy is numerical agreement with two independent references, performed hierarchically, beginning with individual mathematical components and progressing towards complete physical calculations:

- comparison of low-level quantities, such as PMNS matrix elements, kinetic phases and matter potentials;
- comparison of intermediate physical outputs, such as effective mixing angles and eigenvalues in matter and mass-eigenstate weights for solar neutrinos;
- comparison of final, detector-integrated oscillation probabilities.

Since PEANUTS itself only implements the propagation processes associated with solar neutrinos, the comparison against PEANUTS is restricted to that regime: vacuum propagation, the solar MSW effect and production-weighted survival probabilities, and Earth regeneration during night-time observation. **nuSQuIDS**, in contrast, solves the propagation problem independently for both the solar and the atmospheric regimes, and is therefore used to cross-check **tpeanuts** across the full scope of the implementation, including atmospheric-neutrino oscillations for which no PEANUTS reference exists. Because **nuSQuIDS** relies on a different numerical method and implementation, this comparison also exposes discrepancies that could arise from conventions or approximations shared between PEANUTS and **tpeanuts** but absent from **nuSQuIDS**, and is particularly valuable in the matter-dominated regime, where PEANUTS relies on approximations that **nuSQuIDS** does not require.

The tests cover representative energies, baselines, electron densities, mass orderings and neutrino/antineutrino configurations. Whenever possible, numerical agreement is verified to machine precision, with residual differences attributed exclusively to floating-point arithmetic or to the use of different numerical integration strategies; this module-by-module validation provides confidence that the translated implementation faithfully reproduces the physical predictions of the reference codes while benefiting from the flexibility and computational capabilities of PyTorch. Probability conservation is checked through

$$\sum_{\beta} P_{\alpha \rightarrow \beta} = 1 \quad (19)$$

within numerical precision.

3.3 Reproduction of experimental and theoretical results

While comparisons with PEANUTS and **nuSQuIDS** verify the numerical consistency of the implementation, a complete validation also requires demonstrating that the translated framework reproduces established physical phenomena and benchmark results reported in the neutrino-physics literature. These tests provide an implementation-independent assessment of the correctness of the oscillation engine and confirm that the translated software preserves the expected physical behaviour across different oscillation regimes.

The validation suite includes the reproduction of several characteristic signatures of neutrino oscillations, each simultaneously exercising a different part of the implementation and detailed, with figures, in section 5.4:

- the **solar day–night asymmetry** $A_{\text{DN}} = 2(P_{\text{day}} - P_{\text{night}})/(P_{\text{day}} + P_{\text{night}})$, compared with

published measurements (Abe et al., 2016; Aharmim et al., 2005), validating the solar production model, Earth propagation, exposure integration and matter effects together;

- the **MSW resonance**, verifying the position of the resonance, the effective mixing angle in matter and the avoided level crossing predicted by the Mikheyev–Smirnov–Wolfenstein mechanism (Mikheyev & Smirnov, 1985; Wolfenstein, 1978);
- the **adiabatic approximation** (Haxton, 1986; Parke, 1986), checking conservation of the instantaneous matter eigenstate, recovery of the vacuum and high-energy adiabatic limits, and consistency with Landau–Zener non-adiabatic corrections;
- **CP violation and CPT symmetry**, verifying the δ_{CP} -dependent modulation of appearance channels and the CPT relation $P(\nu_\alpha \rightarrow \nu_\beta) = P(\bar{\nu}_\beta \rightarrow \bar{\nu}_\alpha)$ under simultaneous sign reversal of the matter potential and CP phase;
- **atmospheric-neutrino oscillograms** in the $(E, \cos \theta_z)$ plane through the PREM profile (Dziewonski & Anderson, 1981), reproducing the disappearance band, mantle/core matter effects and resonance enhancement reported by atmospheric experiments (Akhmedov et al., 2013);
- **Earth matter tomography**, comparing probabilities under different Earth density models (Gonzalez-Garcia et al., 2008) to verify sensitivity to mantle/core structure;
- the **LMA-Dark degeneracy** (Esteban et al., 2018), reproduced through the simultaneous transformation $\theta_{12} \rightarrow \pi/2 - \theta_{12}$, $\varepsilon_{ee} \rightarrow -(2 + \varepsilon_{ee})$, verifying that the NSI implementation recovers the expected near-degenerate signature.

Together, these benchmarks validate vacuum propagation, matter-enhanced oscillations, solar and atmospheric neutrinos, CP-violating effects and beyond-the-Standard-Model scenarios.

3.4 Benchmarking

Beyond verifying physical correctness, an important objective of this work is to evaluate the computational performance of **tpeanuts** relative to PEANUTS and nuSQuIDS. Since one of the principal motivations for adopting a tensor-based framework is the efficient evaluation of large numbers of oscillation configurations, the benchmarking methodology focuses not only on execution speed but also on scalability and hardware utilisation.

The performance study is based on repeated evaluations of oscillation probabilities under identical physical conditions for the three packages: the same oscillation parameters, density profiles, numerical precision and propagation scenarios are used throughout, so that any performance differences arise exclusively from the numerical implementation rather than from the underlying physics. The benchmark suite explores a broad range of workloads, varying

- the number of neutrino energy points;
- the number of propagation baselines or matter-density sampling points;

- the tensor batch size;
- the computational device (CPU or GPU);
- the floating-point precision (`float32` and `float64`);
- the oscillation scenario (vacuum, matter and solar propagation).

For each configuration, the principal performance metrics are total wall-clock execution time, speed-up relative to PEANUTS and to `nuSQuIDS`, memory consumption, and computational throughput measured as oscillation probability evaluations per second.

Particular emphasis is placed on scalability. Whereas PEANUTS and `nuSQuIDS` are primarily designed to evaluate individual oscillation configurations sequentially, `tpeanuts` performs the same calculations simultaneously through batched tensor operations, so that performance is expected to improve progressively as the batch size increases and the cost of the underlying linear-algebra operations is amortised over many independent evaluations. This is especially relevant for large parameter scans, Monte Carlo simulations and machine-learning applications, where millions of oscillation probabilities may need to be evaluated efficiently.

3.4.1 Benchmarking procedure and CPU/GPU performance

To ensure reproducible, unbiased measurements, all tensors participating in a computation are pre-allocated on the target device, avoiding implicit CPU–GPU transfers; GPU timings are taken only after explicit `torch.cuda.synchronize()`, since CUDA kernels execute asynchronously; and several warm-up iterations, covering context initialisation and kernel compilation, are excluded from the reported, repetition-averaged timings.

Both CPU and GPU hardware are evaluated, in both single- (`float32`) and double-precision (`float64`) arithmetic, the latter being the preferred choice for direct comparison with PEANUTS and `nuSQuIDS`, which perform their reference calculations in double precision. For small workloads, kernel-launch overhead and memory-transfer latency may outweigh the benefits of GPU parallelism, making CPU execution competitive or preferable; as the batch size increases, the tensor-parallel architecture of modern GPUs enables substantially higher throughput.

3.5 Detector-level validation

Validating the detector and inference layers (sections 4.1.6 and 4.1.7) requires a different strategy from the code-to-code and physical-signature comparisons above, since there is no independent reference implementation to compare against: correctness is instead established against real, published experimental data and results. Three complementary checks are applied consistently across every implemented experiment:

- a **closure test**, evaluating the full forward model, oscillation, flux, cross section and detector response, at the experiment’s own published best-fit oscillation parameters with no free normalisation, and comparing the predicted event count or spectrum directly against the real observed data;

- an **absolute-normalisation audit**, when the closure test reveals a systematic offset, tracing it through every stage of the forward-model chain (target mass, efficiency, cross section, source flux, exposure) against independently published values, to establish whether the discrepancy is a known physical effect, a nuisance-parameter freedom the official analysis also carries, or an implementation error;
- a **parameter-recovery fit**, minimising the appropriate likelihood of section 2.4 with respect to the oscillation parameters (and, where relevant, normalisation and background nuisances), and comparing the fitted values, with their Laplace uncertainties, against the experiment’s own published best fit and, independently, against the global-fit values from NuFIT (Esteban et al., 2024).

This methodology is applied in full to Daya Bay, the primary case study of section 5.6, and, in summary form, to Borexino, IceCube DeepCore and SNO phase II (section 5.7).

4 Technical Development

`tpeanuts` is structured as a conservative rewrite of the numerical core of PEANUTS. The implementation follows the original physical decomposition while replacing array operations by PyTorch tensors. This section describes the main components of the update.

4.1 Software architecture

`tpeanuts` is a PyTorch-native reimplementaion of the neutrino oscillation and flux-propagation toolkit originally developed as PEANUTS (a NumPy/Numba package), extended into a full detector-level statistical-inference framework. The guiding design principle is to replace the scalar, per-point evaluation model of the original implementation with a batched, tensor-oriented formulation: every physical quantity, Hamiltonians, evolution operators, oscillation probabilities, fluxes, predicted event counts, likelihoods, is expressed as a function of broadcastable, differentiable `torch.Tensor` inputs, so that arbitrarily large grids of energies, angles and detector configurations can be evaluated, and gradients through them backpropagated, in a single vectorised call. This is the central architectural departure from both the legacy implementation and from reference tools such as `nuSQuIDS` (Arguelles et al., 2022), whose C++/pybind11 API is fundamentally scalar, with one solver object per energy/angle point.

The codebase is organised into nine layers of increasing physical and statistical specialisation, summarised in table 1: a configuration layer centralising model choices, a domain-independent core layer, a set of medium-specific physics modules, a collection of external-tool integrations, a source layer for solar and reactor emission models, a detector layer folding oscillated fluxes into predicted event counts, an inference layer fitting model parameters to real data, a pipeline orchestration layer composing these into end-to-end observables, and a shared utility layer.

Table 1: Software layers of `tpeanuts` and their responsibility.

Layer	Responsibility
<code>config/</code>	Composed configuration objects (<code>PropagationConfig</code> , <code>SolarParameters</code>) and named SM/B-SM/NSI presets, replacing per-call keyword arguments.
<code>core/</code>	Medium-independent Hamiltonians, PMNS/mass-spectrum construction, evolution operators; standard and BSM ($3 + 1$ sterile, NSI, Majorana) variants.
<code>medium/</code>	Environment-specific density/geometry models: vacuum, solar, Earth, atmosphere.
<code>source/</code>	Emission models at the neutrino source: solar production profiles, reactor (Huber–Mueller) antineutrino spectra.
<code>external/</code>	Optional third-party integrations (MCEq, nuSQuIDS, Honda, PyMSIS) for cross-validation and flux/density inputs.
<code>detector/</code>	Forward-folding of oscillated flux into predicted event counts: cross sections, target composition, energy response, efficiency, backgrounds, per experiment.
<code>inference/</code>	Likelihood construction, gradient-based (LBFGS) fitting, Laplace/Fisher uncertainty estimation, likelihood scans.
<code>pipeline/</code>	Composition of the above into complete observables (solar/Earth/atmosphere detection chains).
<code>util/</code>	Device/dtype management (<code>RuntimeContext</code>) and generic numerical primitives.

4.1.1 Configuration layer (`config/`)

`config/propagation.py` defines `PropagationConfig`, a single composed object replacing the 20–38 separate keyword arguments previously threaded through each pipeline call; `config/solar.py` composes it with source-specific parameters into `SolarParameters`. `config/presets.py` is the single registry of named Standard-Model, sterile and NSI parameter presets (e.g. NuFIT best fits, LMA-Dark, published BSM benchmarks) reused consistently across the core engine, the detector layer and the notebooks. `config/default.py` centralises remaining numerical and I/O defaults (density-profile choice, MCEq/PyMSIS backend selection, exposure defaults).

4.1.2 Core layer (`core/`)

The core package implements the medium-independent building blocks of three-flavour, and optional BSM ($3 + 1$ sterile, NSI, Majorana; `core/BSM/`), neutrino oscillation: the vacuum and matter Hamiltonians, PMNS/mass-spectrum construction (`core/common/mass_spectrum.py` defines an abstract `MassSpectrum` base, specialised by `core/SM/` and `core/BSM/`), the matter potential, and the machinery to convert a Hamiltonian into a unitary evolution operator. Two complementary propagation strategies are implemented: a perturbative/analytical evolver (`core/perturbative/`), exploiting a closed-form or series expansion for smooth, slowly varying density profiles, and a medium-independent numerical evolver (`core/numerical/`), which discretises the trajectory, exponentiates a local Hamiltonian on each segment via `torch.linalg.matrix_exp`, and composes the segment operators through an $O(\log N)$ binary-tree reduction rather than a sequential product, keeping the composition step fully batchable and GPU-friendly.

4.1.3 Medium-specific physics (`medium/`)

Each propagation environment specialises the core layer with an environment-specific density/geometry model: `medium/vacuum/`, `medium/solar/` (adiabatic MSW/Landau–Zener mixing with production-point-averaged fluxes), `medium/earth/` (matter regeneration along nadir-angle trajectories, with an exposure module time-averaging over the detector’s latitude and the Earth’s

rotation via incomplete elliptic integrals), and `medium/atmosphere/` (propagation from a production height through the atmosphere and, optionally, into the Earth). Each subpackage follows a consistent `profile.py/evolutor.py/probability.py/flux.py` structure.

4.1.4 Source layer (`source/`)

This layer isolates the physical model of the neutrino *emission* itself, upstream of propagation: `source/solar/model.py` implements the Standard Solar Model production distributions consumed by `medium/solar/`, and `source/reactor/huber_mueller.py` implements the per-isotope Huber–Mueller (Huber, 2011; Mueller et al., 2011) reactor-antineutrino emission spectrum consumed by `detector/dayabay/`.

4.1.5 External-tool integrations (`external/`)

This layer isolates optional third-party dependencies from the core physics code: `external/mceq/` wraps the MCEq cascade-equation solver (Fedynitch et al., 2015); `external/nusquids/` wraps nuSQuIDS (Arguelles et al., 2022) as an independent reference for validation and benchmarking; `external/honda/` reads the Honda/HKKM atmospheric-flux tables (Honda et al., 2007); `external/pymis/` provides an empirically calibrated atmospheric density model (Emmert et al., 2021). Each is imported lazily so the package degrades gracefully when the optional extras are not installed.

4.1.6 Detector layer (`detector/`)

The detector layer folds an oscillated flux, expressed as a survival/transition probability and an incident flux, into a predicted, binned event count observable, entirely decoupled from how the probability was computed. `detector/common/` provides the detector-agnostic building blocks: `target.py` (target electron/nucleon count from stoichiometry), `response.py` (Gaussian energy-response matrix construction), `efficiency.py`, `background.py`, and `event_rate.py::predicted_counts`, the single folding routine $N_k = \text{exposure} \times \sum (\text{flux} \times \text{cross section} \times \text{response} \times \text{efficiency}) + \text{background}$, common to every experiment. `detector/interaction/` implements the relevant cross sections by physical process rather than by detector: inverse beta decay to order $1/M$ (Vogel & Beacom, 1999) (reactor antineutrinos), neutrino–electron elastic scattering (solar), and deuteron dissociation (SNO). On top of these, five experiment-specific subpackages specialise the target, response and background: `detector/borexino/`, `detector/dayabay/`, `detector/icecube/`, `detector/sno/` and `detector/sno_ii/`, summarised in table 2 and, for Daya Bay, described in detail in section 5.6.

4.1.7 Inference layer (`inference/`)

The inference layer implements the statistical formalism of section 2.4 as reusable, detector-agnostic components: `inference/model.py` defines the `DifferentiableModel` protocol (a `free` parameter tuple and a `predict(theta)` method) that every concrete model, oscillation-only or detector-composed, implements; `inference/likelihood.py` provides `poisson_nll`,

Table 2: Experiments implemented on the detector/inference layers, and their validation status against real published data (section 5.7).

Experiment	Channel / detector physics	Status against real data
Daya Bay	Reactor $\bar{\nu}_e$ IBD, 8 detectors, 6 reactors, full flux/response chain	Near/far fit recovers $\sin^2 2\theta_{13}$, Δm_{31}^2 jointly within 1–2% of published (section 5.6); full-shape fit not yet robust
IceCube DeepCore	Atmospheric ν_μ disappearance, event-by-event weighted-MC reweighting	Fit recovers θ_{23} , Δm_{3l}^2 within -0.05σ of published; fitted muon-background normalisation traced to fixed detector systematics
SNO II (salt phase)	CC/NC/ES, 38 published observables, full stat.+syst. covariance	Fit converges to θ_{12} , Δm_{21}^2 , λ_{8B} consistent with SNO’s own SNO-only range; Laplace uncertainty on Δm_{21}^2 breaks down once the full covariance is included, motivating the likelihood-scan machinery
Borexino	Solar ν - e elastic scattering, pp-chain spectrum	Detector response and event-rate chain implemented; background model and energy-resolution normalisation not yet taken from the published analysis
SNO (phase I)	CC/NC/ES, pure-D ₂ O phase	Cross sections, response and inference model implemented and unit-tested; not yet exercised against a full published data set

`chi2_asymmetric`, `correlated_gaussian_nll` (for covariance-matrix observables such as SNO II’s 38-observable data set) and `gaussian_prior_penalty`; `inference/fit.py::fit_lbfgs` performs the LBFGS minimisation and Laplace/Fisher covariance estimate; `inference/scan.py::loglik_grid` evaluates two-dimensional likelihood scans, with an optional per-point re-minimisation of the remaining free parameters. The concrete, experiment-specific models composing an oscillation model with a detector’s event-rate function live alongside each detector, in `detector/<experiment>/inference_model`.

4.1.8 Pipeline and orchestration (pipeline/)

The pipeline package composes the medium-specific modules into complete, physically meaningful observables, a solar-production-to-Earth-detection chain, or a full atmospheric flux propagated through the Earth. A legacy adapter over the bundled `peanuts/` package is retained in full as the ground truth against which the torch reimplementation is validated numerically, cell by cell, throughout `notebooks/validation/legacy/`.

4.1.9 Utilities (util/)

A `RuntimeContext` object centralises device and floating-point-precision resolution, so every physics function accepts an optional context argument instead of separate device/dtype parameters. `util/math.py` supplies torch-native numerical primitives shared across domains: interval intersection, trapezoidal integration, Carlson symmetric elliptic integrals, and the binary-tree matrix-product reduction used by `core/numerical/`.

4.1.10 Validation and performance methodology

Correctness is established by bit-level agreement against the legacy PEANUTS implementation and against `nuSQuIDS`, automated as notebooks (`notebooks/validation/`) for every propagation medium, complemented by the detector-level validation methodology of section 3.5 for the detector/inference layers. Performance is characterised separately (`notebooks/benchmark/`), timing `tpeanuts`’s batched tensor evaluation against both references’ scalar, per-point APIs.

4.2 Numerical simulation conventions

The numerical core of `tpeanuts` is built around a small number of deliberate conventions that keep every propagation medium numerically consistent, dimensionally well-behaved, and directly comparable to both the legacy PEANUTS implementation and to `nuSQuIDS`. The most important of these are the choice of a unitary, dimensionless evolution scale for the Hamiltonian, the neutrino/antineutrino sign convention, and the floating-point precision policy used throughout the matrix-exponential evolution.

4.2.1 Non-dimensionalisation of the Hamiltonian

Rather than working directly with the physical matter potential and kinetic (mass-squared) terms, which span many orders of magnitude depending on the neutrino energy and the medium’s density, every Hamiltonian contribution is first converted into a dimensionless quantity by multiplying it by a configurable evolution length scale, `evolution_scale_m` (`util/constant.py::R_E`, the Earth radius, by default). Concretely,

$$V = s L_{\text{scale}} \sqrt{2} G_F n_e, \quad k_i = L_{\text{scale}} \frac{\Delta m_i^2}{2E\hbar c}, \quad (20)$$

where $s = \pm 1$ is the neutrino/antineutrino sign (see below), n_e is the electron number density, and Δm_i^2 are the (reduced) mass-squared eigenvalues. The trajectory coordinate is normalised by the same scale, $x = L/L_{\text{scale}}$, so that the evolution operator is always computed as $U = \exp(-iHx)$, with both H and x dimensionless.

This convention has two practical benefits for a batched, GPU-oriented implementation: it keeps the arguments passed to `torch.linalg.matrix_exp` at order-unity magnitude regardless of the physical baseline, from a few kilometres for the atmosphere to $\sim 7 \times 10^5$ km for solar propagation, which improves the numerical conditioning of the matrix exponential and avoids overflow/underflow when many disparate energies and baselines are evaluated in the same batched call; and it decouples the physical unit system from the propagation length, since only the ratio L/L_{scale} ever enters the exponent. The evolution scale defaults to the Earth’s radius purely for backward compatibility with the legacy PEANUTS convention, but it is an explicit, per-medium parameter and can be set independently, e.g. to the injection height of an atmospheric shower rather than to $R_{\oplus}$.

4.2.2 Neutrino/antineutrino convention

Antineutrino propagation is selected through a single boolean (or boolean-tensor, for batched mixed samples) flag, `antinu`, carried by the `OscillationParameters` object and threaded through every Hamiltonian and probability routine. Two independent sign flips are applied consistently whenever `antinu=True`:

- **Matter potential.** The Wolfenstein charged-current potential changes sign, $V \rightarrow -V$, since it originates from the coupling of ν_e (not $\bar{\nu}_e$) to the ambient electrons.
- **Mixing matrix.** The PMNS matrix is replaced by its complex conjugate, $U \rightarrow U^*$, equivalent to flipping the sign of the CP-violating phase, $\delta_{\text{CP}} \rightarrow -\delta_{\text{CP}}$.

Both substitutions are applied at the point where the reduced mixing matrix and the matter Hamiltonian are assembled (`core/common/pmns.py::select_antinu`, `core/common/potential.py::matter_p`) so that every downstream propagation routine, vacuum, solar, Earth or atmospheric, inherits the correct antineutrino behaviour automatically from a single flag, without medium-specific antineutrino logic. The same object also encodes the neutrino mass ordering as the sign of the atmospheric splitting Δm_{3l}^2 : a positive value selects normal ordering and a negative value selects inverted ordering, both handled by a single `torch.where` branch, so that mixed-ordering batches (e.g. for an ordering-marginalised analysis) can, in principle, be evaluated in one call.

4.2.3 Precision and evaluation mode

All propagation is carried out at double precision (`torch.float64` real / `torch.complex128` complex) by default, matching the precision of the legacy NumPy implementation and keeping the matrix-exponential composition numerically stable over trajectories segmented into hundreds of steps. Every core routine is wrapped in `@torch.no_grad()`: the package is designed for forward physical evaluation, not for gradient-based fitting, so autograd bookkeeping is disabled throughout to reduce memory footprint and computational overhead.

For validation purposes, a `legacy_precision` flag is threaded through the potential and Hamiltonian builders; when enabled, it substitutes the full-precision matter-potential prefactor, derived from CODATA G_F , N_A and $\hbar c$ (Tiesinga et al., 2021), with the four-significant-digit constant hard-coded in the original PEANUTS implementation, allowing bit-level, rather than merely physics-level, agreement to be verified against the legacy reference where required.

4.3 Neutrino detection pipelines

Beyond the medium-specific propagation primitives described above, the `pipeline/` package assembles them into two complete, physically distinct detection workflows: a solar-neutrino pipeline, which propagates neutrinos produced deep inside the Sun to a terrestrial detector, and an atmospheric-neutrino pipeline, which propagates neutrinos produced by cosmic-ray air showers in the Earth’s atmosphere down to a detector. Both share the same underlying core Hamiltonian/evolutor machinery, but differ fundamentally in one respect: whether the mass-eigenstate coherence acquired at production survives all the way to the detector.

4.3.1 Solar-neutrino detection pipeline

A solar neutrino is produced as a pure ν_e at some radius ρ inside the Sun, distributed according to a source-specific production profile (e.g. ^8B , pp ; `medium/solar/profile.py`, `io.py`) (Vinyoles et al., 2017). As it propagates outward through the Sun’s smoothly varying electron density, the MSW matter effect adiabatically rotates the flavour state into the local mass-eigenstate basis (`medium/solar/matter_mixing.py`), with `medium/solar/landau_zener.py` providing the correction for the, generally small, probability of a non-adiabatic level-crossing jump between eigenstates (Parke, 1986). Because the $\sim 1.5 \times 10^8$ km vacuum baseline to Earth is enormously larger than the neutrino’s oscillation length at solar-neutrino energies, the relative phases between mass eigenstates wash out completely before arrival: only the incoherent mass-eigenstate populations survive, not a coherent superposition.

`tpeanuts` implements this picture in `pipeline/solar_earth.py`, which reproduces, in batched `torch` form, the original PEANUTS NumPy/Numba reference workflow (`pipeline/legacy/`) used to validate it bit-for-bit. The pipeline proceeds in three stages:

1. **Solar production and adiabatic conversion** (`solar_probability_mass`): for a given source and energy grid, the electron-neutrino state is propagated adiabatically outward from each production radius ρ in the source’s normalised production-radius distribution, and the result is averaged over that distribution to yield a single vector of mass-eigenstate weights, $w_i(E) = (w_1, w_2, w_3)$, the fraction of the flux exiting the Sun in each mass eigenstate.
2. **Vacuum propagation to Earth**: because no relative phase is ever tracked, propagation across the Sun–Earth vacuum baseline leaves the mass weights $w_i(E)$ unchanged by construction; this step is a physical no-op in the incoherent formulation, not an approximation applied afterwards.
3. **Earth matter propagation** (`pearth(..., massbasis=True)`): at the detector, the incoherent mixture of incident mass eigenstates is propagated through Earth matter, relevant only for underground detectors sensitive to the day/night matter-regeneration asymmetry, evaluated over a grid of nadir angles η , giving the final flavour-transition probability $P_{\nu_e \rightarrow \nu_i}(E, \eta)$.

The final detector-level probability is obtained as a triple average: over the solar production-radius distribution (folded into the first stage), over the neutrino energy spectrum, and, for underground detectors, over the detector’s nadir-angle exposure, the fraction of a year during which each nadir angle is observed, computed from the detector’s geographic latitude and the Earth’s daily and annual motion (`medium/earth/exposure_table.py`, `exposure_math.py`) and folded in via `integrate_exposure`. The pipeline additionally accepts a `legacy_precision` flag that substitutes the full-precision matter-mixing prefactor with the legacy PEANUTS hard-coded constant, enabling bit-comparable validation against the original reference implementation rather than mere physics-level agreement.

4.3.2 Atmospheric-neutrino detection pipeline

Atmospheric neutrinos follow a different physical picture, implemented in `pipeline/atmosphere.py` and `pipeline/atmosphere_earth.py`. They are produced not at a single well-defined point but across a broad, energy- and angle-dependent distribution of production heights h in the atmosphere, with the boundary-condition flux $\Phi_\beta(E, h, \theta)$ for each produced flavour β supplied by an external cascade-equation generator (MCEq Fedynitch et al., 2015) or by tabulated Honda/HKKM fluxes (Honda et al., 2007). Crucially, the production height is only a few tens of kilometres from the detector, many orders of magnitude shorter than the solar baseline, so coherence is never lost: the full flavour-amplitude state is propagated coherently from production to detector. The pipeline proceeds in five stages:

1. **Flux selection:** one produced particle and one detector zenith angle are sliced out of the loaded height/energy/angle flux dataset.
2. **Atmosphere propagation:** a coherent unit-amplitude flavour state $|\beta\rangle$ is evolved from each production height down to the Earth’s surface, batched jointly over the full energy $\times$ height grid.
3. **Earth propagation:** for an upward-going, Earth-crossing trajectory, the surface state is propagated further through Earth matter to the detector depth, using the same coherent evolver as the solar pipeline’s coherent variant.
4. **Height integration:** the detector-level flux for each final flavour is obtained by weighting the propagated $\beta \rightarrow i$ probability at each height by $\Phi_\beta(E, h, \theta)$ and trapezoidally integrating over h .
5. **Flavour summation:** the height-integrated contributions from every produced flavour (ν_e, ν_μ, ν_τ and antiparticles) are summed into the total detector-level flux of each final flavour.

4.4 Test suite and sanity checks

Correctness in `tpeanuts` is established through a dedicated `pytest` suite of 846 test functions organised into 64 files that mirror the package’s module tree one-to-one: `core/`, `medium/` and `source/` test directories for the physics engine, as before, plus new `detector/common/test/`, `detector/interaction/test/` and one `test/` directory per experiment, and `inference/test/`, added as the detector and inference layers were built. The suite is run automatically whenever a module is modified. The tests fall into four broad, partly overlapping categories.

4.4.1 Code consistency checks

These verify the software contract of each function independently of the underlying physics: expected output shape/dtype/device; correct broadcasting over batched grids; correct exceptions on invalid inputs; mutual consistency between alternative code paths (e.g. “analytical” vs. “numerical”); and save/load round-trips reproducing the original data. This is the largest category, roughly a third to two-fifths of the suite.

4.4.2 Mathematical checks

These verify that a function’s output matches an independently derived closed-form expression or exact algebraic identity: the PMNS matrix’s unit determinant and Jarlskog invariant; the composed segment-evolutor product against a manually accumulated matrix product; agreement between the perturbative and purely numerical evolutors on a constant-density segment; a resonance radius against its closed-form solution; and every non-dimensionalised term against its defining formula under parameter scaling. Roughly a quarter to a third of the suite.

4.4.3 Physical coherence checks

These verify that computed quantities respect the physical constraints of the theory, independently of how they were computed: probability conservation (every transition-probability matrix doubly stochastic); unitarity of every evolution operator; recovery of known physical limits (zero baseline/density reduces to the identity, the adiabatic solar limit as the Landau–Zener probability vanishes, the expected antineutrino day/night asymmetry direction); and, most stringently, function-by-function cross-validation against the original PEANUTS NumPy/Numba reference. Roughly a third to two-fifths of the suite.

4.4.4 Detector and inference validation

Because the detector and inference layers have no independent reference implementation to compare against, their `pytest` coverage (`detector/*/test/`, `inference/test/`) focuses on code-consistency and mathematical checks, e.g. the exact invertibility of `DayaBayExperimentalModel`’s $(\theta_{13}, \Delta m_{31}^2) \leftrightarrow (\sin^2 2\theta_{13}, \Delta m_{ee}^2)$ reparametrisation, or that the $\text{IAV} \rightarrow \text{LSNL} \rightarrow \text{Gaussian}$ -resolution response composition conserves total probability, while the physical-coherence role played elsewhere by comparison with PEANUTS/`nuSQuIDS` is instead played by the closure tests and parameter-recovery fits of section 3.5, run as notebooks rather than as `pytest` cases.

4.4.5 Notebook-level validation

Beyond the `pytest` suite, a second, complementary layer of validation lives in the `notebooks/validation/`, `notebooks/diagnostic/` and `notebooks/inference/` trees: dedicated notebooks reproduce bit-level agreement against legacy PEANUTS and against `nuSQuIDS` (Arguelles et al., 2022) for every propagation medium, run the detector-level closure tests and fits of section 3.5 for each experiment, and embed inline “smoke test” cells, printed diagnostics followed by `assert` statements, that must pass before the notebook is considered valid.

5 Validation and Results

This section presents the validation outputs, benchmark results and pipeline-level analyses obtained for `tpeanuts`, following the methodology described in sections 3.2, 3.3 and 3.5. The chapter is organised into seven parts: numerical validation against PEANUTS and `nuSQuIDS`, performance benchmarking against both references, an analysis of the solar and atmospheric

detection pipelines, the reproduction of well-known experimental and theoretical results, the BSM extensions (NSI and sterile neutrinos), a detector-level case study on real Daya Bay data, and a summary of the remaining detector-inference applications.

5.1 Validation

5.1.1 tpeanuts vs PEANUTS

Bit-level agreement with the legacy PEANUTS implementation is the primary correctness criterion for the translation (section 3.2). Figure 1 summarises the maximum absolute and relative differences observed across the four validated modules, solar, Earth, vacuum and the composed detection pipeline, when both codes are evaluated with the `legacy_precision` flag enabled (section 4.2.3). The solar and Earth modules agree to within 10^{-14} – 10^{-9} in relative terms, well below the 1 ppb reference line; the vacuum module agrees to $\sim 10^{-7}$; and the composed pipeline, which accumulates rounding from every intermediate stage, agrees to a few parts in 10^5 , marginally crossing the 1 ppm reference line while remaining orders of magnitude below any physically meaningful tolerance.

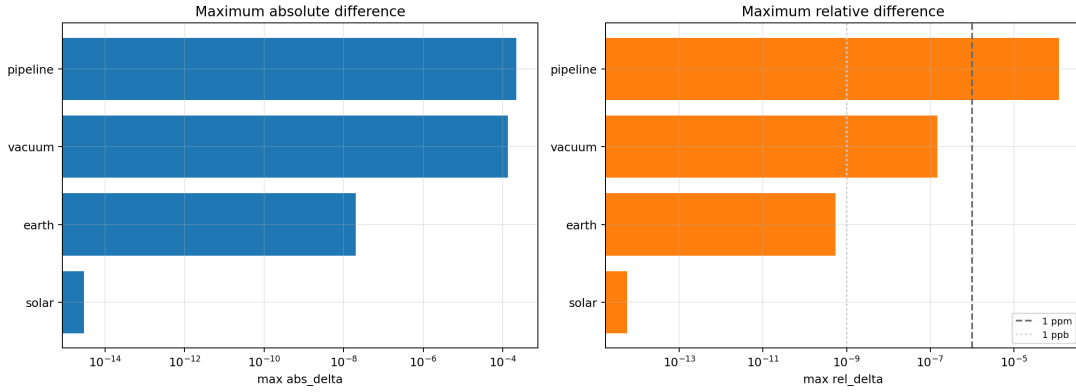


Figure 1: Maximum absolute and relative differences between `tpeanuts` and legacy PEANUTS, by validated module (solar, Earth, vacuum, pipeline). Dashed and dotted lines mark the 1 ppm and 1 ppb reference levels.

Isolating the origin of this residual pipeline-level discrepancy for the solar detection pipeline (^8B , $\eta = 0.75$ rad): comparing the `full_precision` evaluation against the `legacy` reference gives an absolute/relative error of 10^{-5} – 10^{-4} , whereas enabling `legacy_precision` reduces the discrepancy by roughly nine orders of magnitude, to 10^{-10} – 10^{-9} . This confirms that the residual difference between `tpeanuts` and PEANUTS is entirely attributable to the updated CODATA-derived constants used in the full-precision matter potential (Tiesinga et al., 2021), rather than to any implementation error.

5.1.2 tpeanuts vs nuSQuIDS

Figure 2 summarises the maximum absolute and relative differences between `tpeanuts` and `nuSQuIDS` across four validated configurations: vacuum, atmosphere, solar and the composed solar-to-Earth-detection pipeline. The vacuum and atmosphere modules agree to within 10^{-8} – 10^{-7} in relative terms, comfortably below the 1 ppb reference line, as expected for two independent,

essentially exact solvers of the same closed propagation problem. The solar module and the composed solar-Earth-detection pipeline instead show markedly larger discrepancies, at the percent ($\sim 10^{-2}$) and $\sim 10\%$ level respectively, several orders of magnitude above the vacuum/atmosphere agreement.

This pattern is consistent with the validation strategy of section 3.2: unlike the vacuum and atmosphere cases, the solar and solar-Earth-detection observables depend on production-averaged, incoherent mass-eigenstate weights and on the treatment of coherence loss between the Sun and the Earth, both of which **tpeanuts** inherits directly from the PEANUTS convention (section 2.1) but which **nuSQuIDS** may implement through different internal approximations. The comparison against **nuSQuIDS** therefore isolates exactly the kind of convention-dependent discrepancy that a comparison against PEANUTS alone, which shares the same conventions, cannot expose.

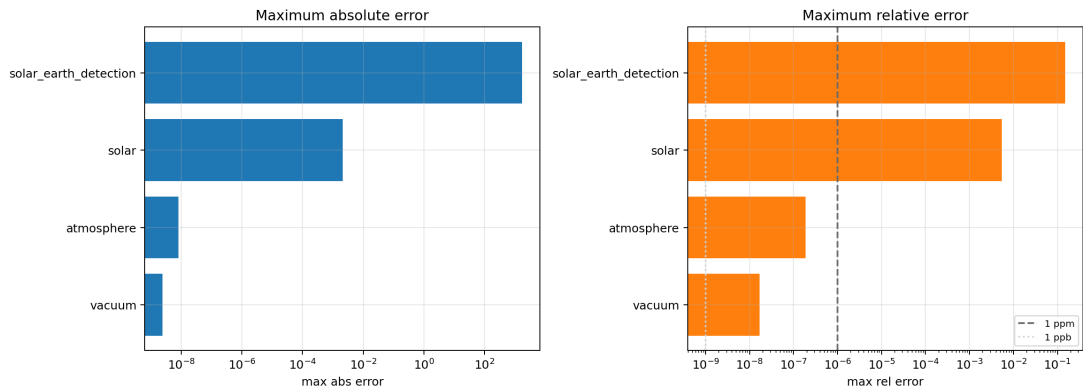


Figure 2: Maximum absolute and relative differences between **tpeanuts** and **nuSQuIDS**, by validated configuration (vacuum, atmosphere, solar, solar-Earth-detection pipeline). Dashed and dotted lines mark the 1 ppm and 1 ppb reference levels.

5.1.3 Internal validation

Beyond comparisons against external reference codes, the perturbative/analytical and the purely numerical (**matrix_exp**-based) evolvers of section 2.1 solve the same propagation problem through independent numerical strategies and must therefore agree with each other. For Earth propagation of an initial ν_e state ($\eta = 0.75$ rad) across the multi-GeV atmospheric range, the two strategies agree with a relative error below $\sim 10^{-2}$ even near the matter-resonance feature at 2–3 GeV, with no systematic drift, a substantially looser but expected tolerance given the many independent segment-boundary roundings accumulated when discretising the non-constant PREM profile, compared to the machine-precision agreement obtained on a single constant-density segment (section 3.2).

5.2 Benchmark

5.2.1 tpeanuts vs PEANUTS

Figure 3 shows the wall-clock speed-up of **tpeanuts** over legacy PEANUTS for the Earth-propagation probability calculation, as a function of the energy-grid and nadir-grid sizes used in the batched evaluation. Unlike the comparison against **nuSQuIDS** (figure 4), the speed-up

here is far more modest: for small grids **tpeanuts** is in fact slower than PEANUTS, by up to a factor ~ 10 at the smallest grid considered (4×4), and only becomes faster once the grid exceeds ~ 32 – 64 points per dimension, reaching $\sim 3.6\times$ at the largest grid (256×1024). This is the expected outcome of comparing against an already vectorised NumPy implementation rather than against a scalar, per-point solver: PEANUTS batches efficiently over array operations for a single trajectory, so the advantage of **tpeanuts**’s tensor-based formulation only emerges once the batch size is large enough for the fixed overhead of constructing and dispatching PyTorch operations to be amortised, consistent with the benchmarking methodology of section 3.2.

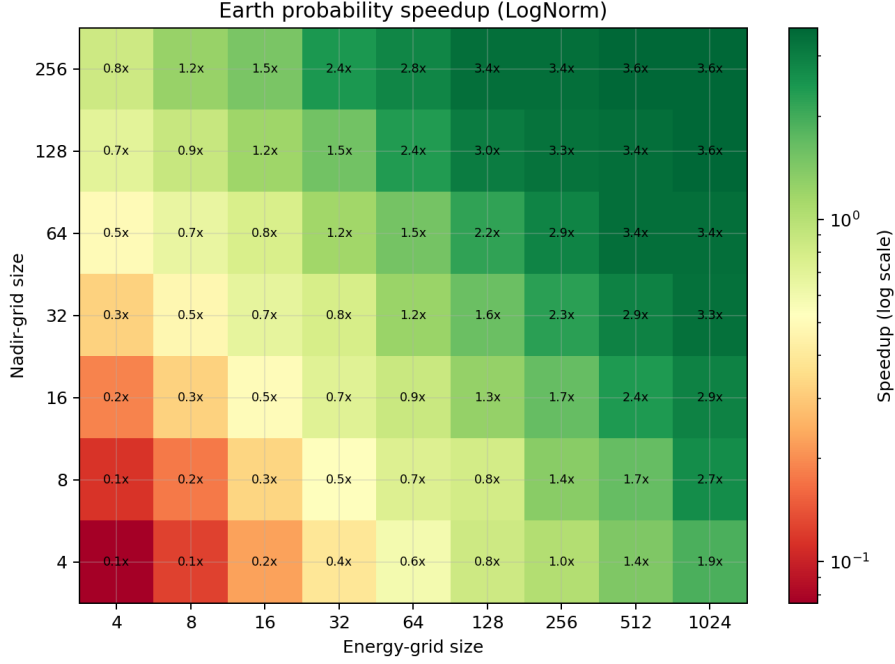


Figure 3: Wall-clock speed-up of **tpeanuts** over legacy PEANUTS for the Earth-propagation probability, as a function of energy-grid and nadir-grid size.

5.2.2 tpeanuts vs nuSQuIDS

Figure 4 shows the wall-clock speed-up of **tpeanuts** over **nuSQuIDS** for the integrated solar-detector probability calculation, as a function of the energy-grid and nadir/ $\cos \theta_z$ -grid sizes used in the batched evaluation. The speed-up grows monotonically with grid size, from $\sim 63\times$ for the smallest grid considered (2×2) to over $1100\times$ for the largest (32×32), directly confirming the scaling behaviour anticipated in section 3.2: because **nuSQuIDS** evaluates one solver object per energy/angle point while **tpeanuts** evaluates the full grid in a single batched call, the relative advantage of the tensor-based formulation increases with the number of configurations evaluated per call.

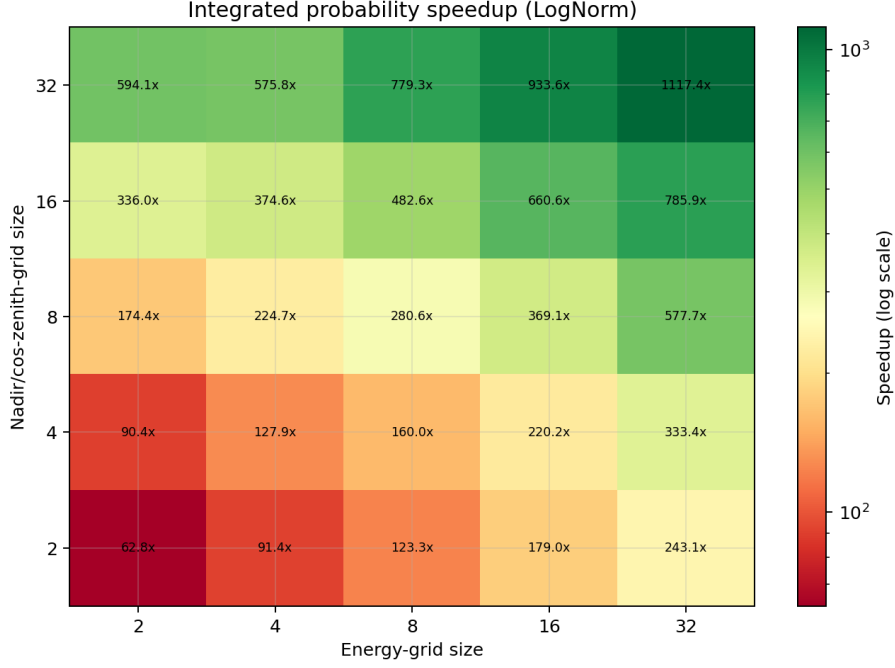


Figure 4: Wall-clock speed-up of `tpeanuts` over `nuSQuIDS` for the integrated solar-detector probability, as a function of energy-grid and nadir-grid size.

5.3 Pipeline analysis

5.3.1 Solar neutrino

Figure 5 exercises the solar detection pipeline (section 4.3) with all solar production channels combined (pp , pep , hep , ${}^7\text{Be}$, ${}^8\text{B}$, ${}^{13}\text{N}$, ${}^{15}\text{O}$, ${}^{17}\text{F}$), giving the total flavour flux $\Phi_\alpha^{\text{tot}}(E, \eta_{\text{fix}})$ at a fixed nadir angle. The ν_e flux falls steadily from $\sim 3.5 \times 10^{10}$ to $\sim 2 \times 10^{10} \text{ cm}^{-2} \text{ s}^{-1}$ as the survival probability decreases with energy, while ν_μ and ν_τ grow correspondingly and converge above $\sim 9 \text{ MeV}$, consistent with the near-equal μ/τ appearance implied by the PMNS structure in the solar sector.

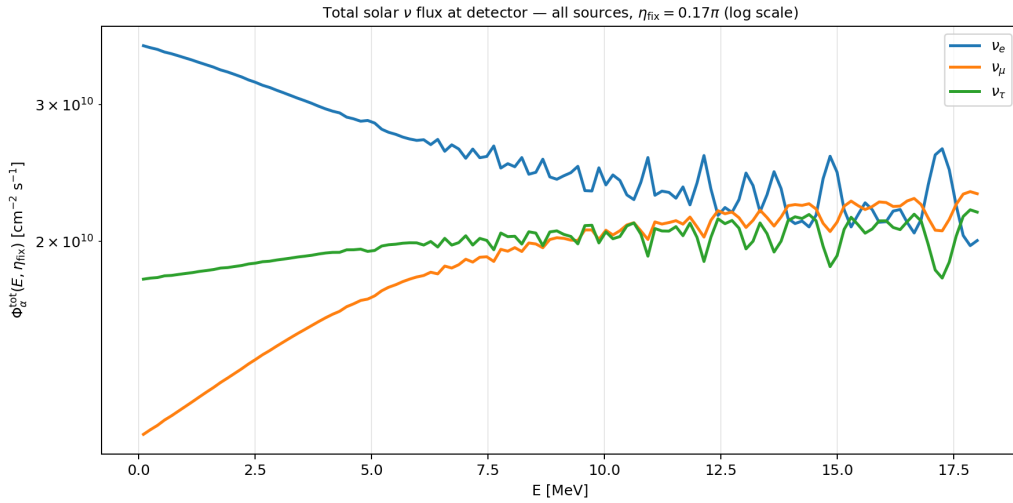


Figure 5: Total solar-neutrino flux at the detector, $\Phi_\alpha^{\text{tot}}(E, \eta_{\text{fix}})$, summed over all solar production channels, at fixed nadir angle $\eta_{\text{fix}} = 0.17\pi$.

Isolating the ^8B channel and exercising the nadir-angle exposure integration (section 4.3) for four detector sites (Super-K, Borexino, JUNO, SNO) gives three flavour spectra that are, to the resolution of the plot, indistinguishable across all four sites: annual averaging over the day/night nadir-angle exposure smooths out the latitude-dependent matter effects seen in figure 5, as expected from the small latitude differences among these detectors.

5.3.2 Atmospheric neutrino

The atmospheric detection pipeline (section 4.3) requires, as a boundary condition, the production-height distribution of the parent flux at each energy and angle. Figure 6 compares this distribution, $f(h|E, \alpha)$, for ν_μ at a fixed zenith angle ($\alpha = 31.8^\circ$) as generated by MCEq (Fedynitch et al., 2015) and by the tabulated Honda/HKKM parametrisation (Honda et al., 2007). Both place the bulk of the production between ~ 10 and 30 km, peaking around 15 km and shifting slightly higher at higher energy; the visibly discretised Honda distribution reflects its coarser production-height binning, but the two agree closely in shape and normalisation.

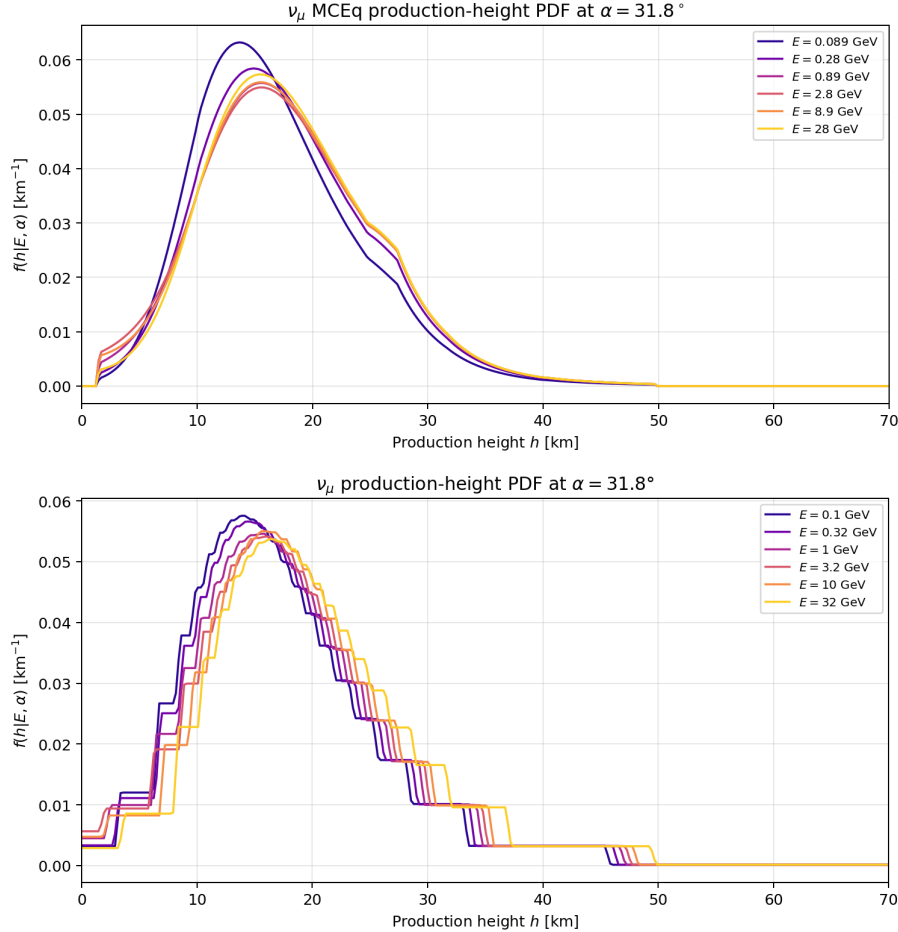


Figure 6: ν_μ production-height probability density, $f(h|E, \alpha)$, at fixed zenith angle $\alpha = 31.8^\circ$, generated with MCEq (top) (Fedynitch et al., 2015) and with the Honda/HKKM parametrisation (bottom) (Honda et al., 2007).

Figure 7 shows the resulting detector-level ν_μ survival probability, $P_{\text{det}}(\nu_\mu \rightarrow \nu_\mu; E, \theta)$, after propagation through the atmosphere and Earth and averaging over the production-height

distribution of figure 6. The pipeline reproduces the expected atmospheric ν_μ disappearance: survival close to unity at high energy ($E \gtrsim 100$ GeV), decreasing towards ~ 0.1 – 0.2 at sub-GeV energies and large zenith angles, confirming that the height-integration and flavour-summation stages combine correctly with the oscillation engine.

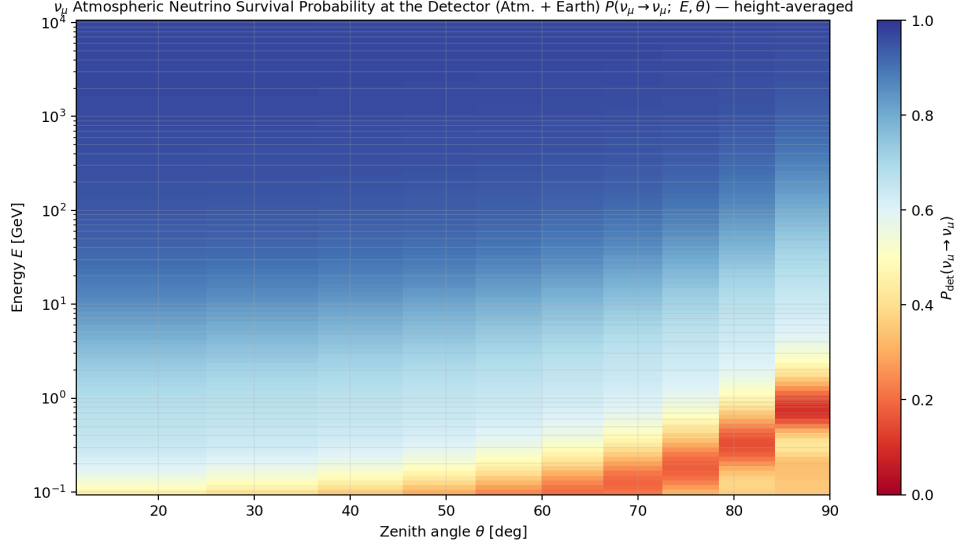


Figure 7: Height-averaged, detector-level ν_μ survival probability $P_{\text{det}}(\nu_\mu \rightarrow \nu_\mu; E, \theta)$, combining atmospheric and Earth propagation.

5.4 Reproduction of experimental and theoretical results

5.4.1 Solar day–night asymmetry

The regeneration of solar ν_e inside the Earth during night-time observation is quantified by the day–night asymmetry $A_{\text{DN}} = 2(P_{ee}^{\text{day}} - P_{ee}^{\text{night}})/(P_{ee}^{\text{day}} + P_{ee}^{\text{night}})$, computed here for the ^8B flux using the NuFIT 5.2 normal-ordering parameters (Esteban et al., 2022). For representative trajectories crossing the inner core, outer core, mantle and crust only, $A_{\text{DN}}(E)$ develops the oscillatory structure characteristic of Earth matter regeneration above ~ 6 MeV, largest for the core-crossing trajectory, consistent with its longer matter path length and higher electron density. Figure 8 compares the flux-weighted, energy-integrated asymmetry $A_{\text{DN}}^{\text{int}}$ obtained for four representative detector sites (Super-K, Borexino, SNO+, JUNO) against the Super-Kamiokande-IV measurement: all four predictions, -2.6% to -2.85% , fall within the experimental 1σ band, $(-3.2 \pm 1.2)\%$, confirming that the solar detection pipeline reproduces the observed regeneration signal within present experimental precision.

5.4.2 MSW resonance

The MSW resonance condition, $V_k(\rho) = \cos 2\theta_{12}$, is located along the solar density profile for several neutrino energies. At low energy ($E \lesssim 1.5$ MeV) the potential never reaches the resonance condition within the Sun, so pp -like neutrinos propagate quasi-adiabatically without crossing resonance; as the energy increases, the resonance radius ρ_{res} moves outward, from $\rho_{\text{res}} = 0.032$ at 2 MeV to $\rho_{\text{res}} = 0.282$ at 15 MeV, reproducing the expected energy dependence of the MSW

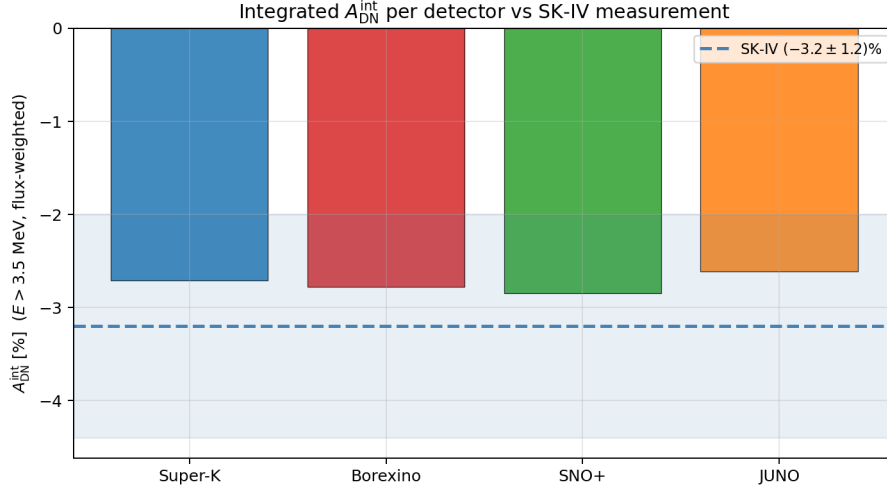


Figure 8: Flux-weighted, energy-integrated day–night asymmetry $A_{\text{DN}}^{\text{int}}$ ($E > 3.5$ MeV) for four representative detector sites, compared with the Super-Kamiokande-IV measurement (Abe et al., 2016).

resonance layer. Figure 9 shows the local survival probability $P_{ee}(\rho, E)$ across the production region. At low energy, P_{ee} remains close to the vacuum-averaged value, $\sum_i |U_{ei}|^4 = 0.553$, throughout the Sun; at high energy, P_{ee} starts near the adiabatic matter-eigenstate limit, $|U_{e2}|^2 = 0.296$, at the production point and relaxes smoothly to the vacuum-averaged value as the neutrino exits the Sun, exactly as expected from adiabatic MSW conversion.

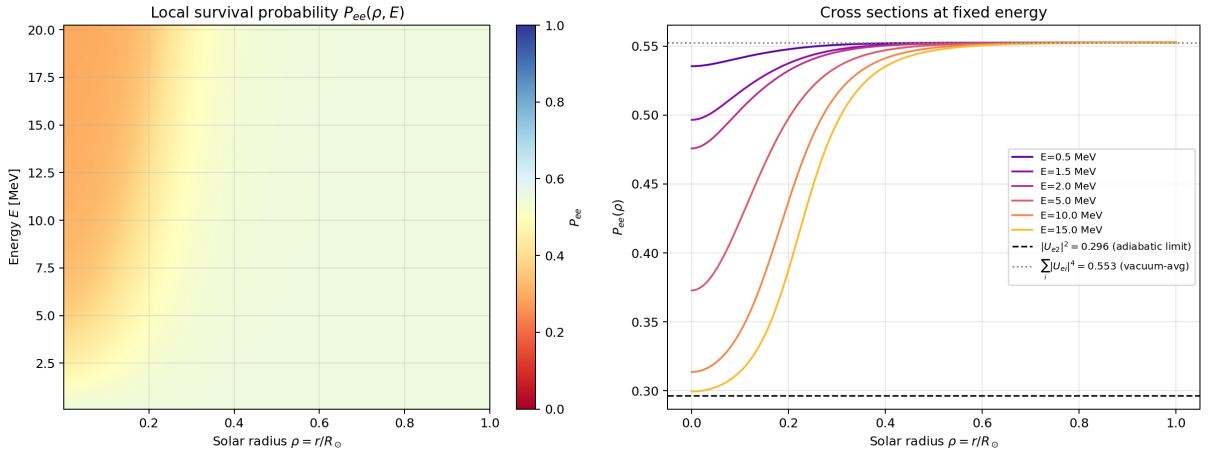


Figure 9: Local survival probability $P_{ee}(\rho, E)$ across the solar production region (left), and radial cross-sections at fixed energy (right), showing the transition between the adiabatic matter-eigenstate limit and the vacuum-averaged value.

5.4.3 Adiabatic approximation

The validity of the adiabatic approximation is assessed through the adiabaticity parameter $\gamma(\rho)$, whose value must remain far above the non-adiabatic onset, $\gamma = 1$, for the adiabatic MSW formalism to apply (Haxton, 1986; Parke, 1986). Across the solar production region and representative energies up to $E = 15$ MeV (the least favourable case), γ remains above $\sim 2 \times 10^3$, several orders of magnitude above threshold, confirming that the adiabatic approximation is an excellent description across the full solar-neutrino energy range and that the Landau–Zener

corrections of section 3.2 remain small.

5.4.4 CP violation and CPT symmetry

Figure 10 shows the appearance probabilities $P_{\mu e}$ and $\bar{P}_{\mu e}$ at a DUNE-like baseline for the NuFIT 5.2 best-fit CP phase, $\delta_{\text{CP}} = 197^\circ$, and for maximal CP violation, $\delta_{\text{CP}} = -90^\circ$, together with the CP asymmetry $A_{\text{CP}} = P_{\mu e} - \bar{P}_{\mu e}$. The asymmetry is largest at low energy, where the genuinely CP-odd term dominates, and vanishes at the first oscillation node near $E \sim 1$ GeV, consistent with the expected L/E dependence.

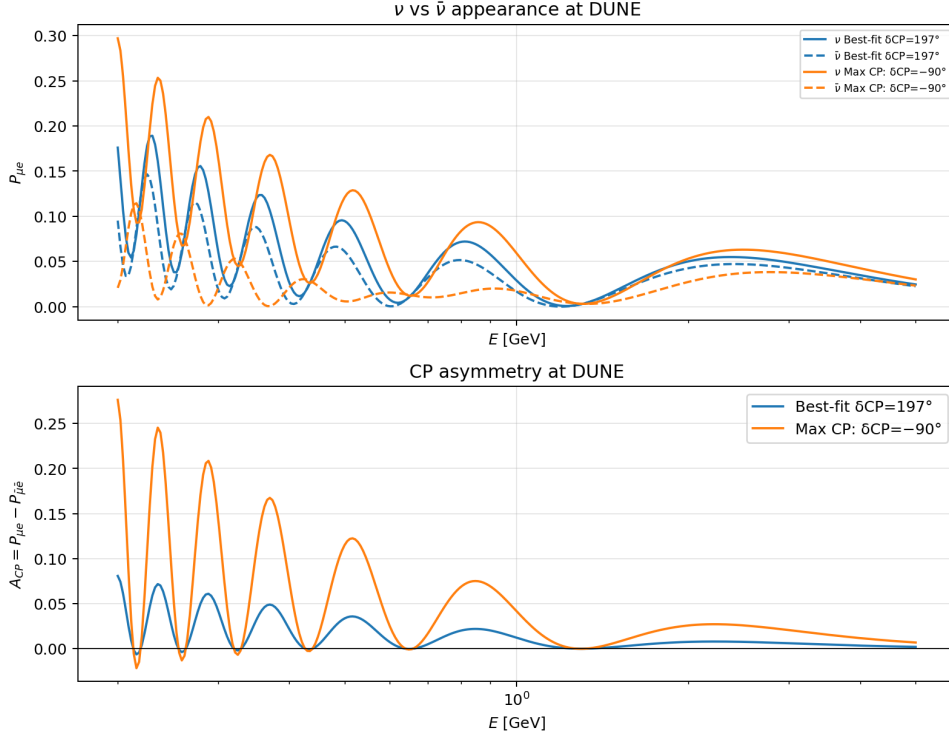


Figure 10: Appearance probabilities $P_{\mu e}$ and $\bar{P}_{\mu e}$ at a DUNE-like baseline (top) and the resulting CP asymmetry A_{CP} (bottom), for the best-fit and maximal-CP-violation values of δ_{CP} .

As an independent check of CPT invariance, $P(\nu_\mu \rightarrow \nu_e)$ is compared against $P(\bar{\nu}_e \rightarrow \bar{\nu}_\mu)$ over a broad grid of oscillation parameters and kinematic configurations; the two probabilities agree exactly over their full range, 0 to ~ 0.45 , confirming that the implementation satisfies the CPT relation $P(\nu_\alpha \rightarrow \nu_\beta) = P(\bar{\nu}_\beta \rightarrow \bar{\nu}_\alpha)$ to numerical precision.

5.4.5 Atmospheric neutrino oscillograms

Figure 11 shows the atmospheric ν_μ survival-probability oscillogram in the $(\cos\theta_z, E)$ plane, computed through the PREM Earth density profile (Dziewonski & Anderson, 1981). The characteristic vacuum oscillation dip is verified against the analytic first-minimum condition, $1.267|\Delta m_{3\ell}^2|L/E = \pi/2$ (Ashie et al., 2004), and against a set of benchmark observables summarised in table 3, spanning the atmospheric mass splitting and mixing angle, the vacuum L/E dip position, the multi-GeV up/down asymmetry, the energy of maximal mantle-crossing suppression, and the maximum appearance probability associated with the mantle MSW resonance.

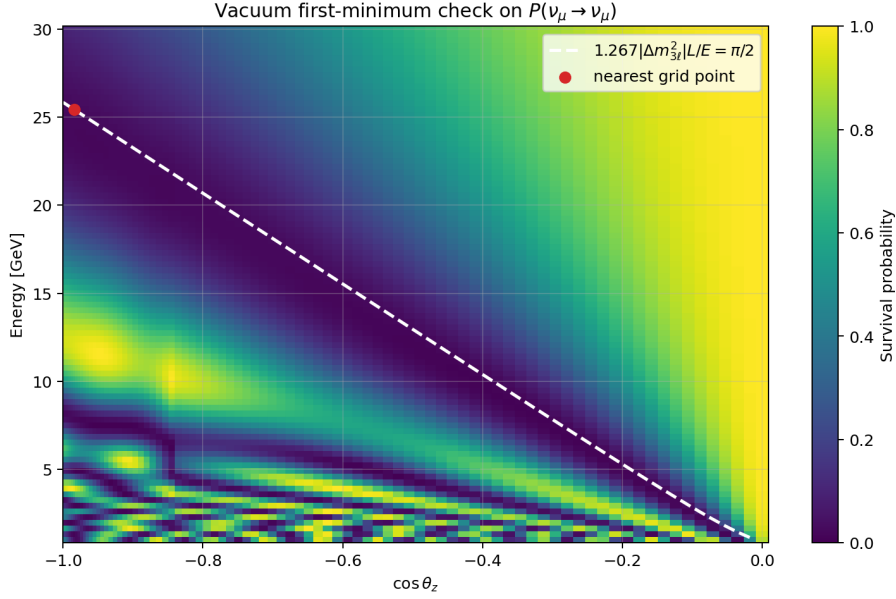


Figure 11: Atmospheric $P(\nu_\mu \rightarrow \nu_\mu)$ oscillogram in the $(\cos \theta_z, E)$ plane. The dashed line marks the analytic first vacuum-oscillation minimum, $1.267|\Delta m_{3\ell}^2|L/E = \pi/2$; the red marker shows the nearest grid point.

Table 3: Benchmark observables reproduced from the atmospheric oscillogram, compared with published experimental results.

Observable	tpeanuts	Experiment	Reference
$ \Delta m_{3\ell}^2 $ [eV ²]	2.511×10^{-3}	$[1.9, 3.0] \times 10^{-3}$	Wester et al. (2024), 90% CL
$\sin^2(2\theta_{23})$	0.981	> 0.90	Wester et al. (2024), 90% CL
L/E first dip	493.7	≈ 500	Ashie et al. (2004)
$A_\mu = (\langle P \rangle_\uparrow - 1)/(\langle P \rangle_\uparrow + 1)$ (uniform flux)	-0.414	≈ -0.32	Fukuda et al. (1998) (Fukuda et al., 1998)
Max-suppression E at $\cos \theta_z = -1$ [GeV]	3.28	3–30 GeV range	Aartsen et al. (2013), DeepCore
max $P(\nu_\mu \rightarrow \nu_e)$ over the grid	0.5904	mantle MSW resonance	Aiello et al. (2024), KM3NeT/ORCA

5.4.6 Earth matter tomography

The derivative of the electron-neutrino appearance probability with respect to the nadir angle, $|dP_e/d\eta|$, develops sharp, localised features precisely at the PREM shell boundaries (Dziewonski & Anderson, 1981), a pronounced spike at the core–mantle density jump and a regular oscillatory pattern for purely mantle-crossing trajectories, confirming that the propagation correctly resolves discontinuous density profiles, the basic requirement for neutrino tomography.

5.4.7 LMA-Dark degeneracy

Figure 12 reproduces the LMA-Dark degeneracy discussed in section 3.2, comparing the day–night asymmetry predicted at fixed nadir angle ($\eta = 30^\circ$, 1 km depth) under three scenarios: the standard LMA solution ($\varepsilon_{ee} = 0$); a sign-flip demonstration in which only the Earth-matter NSI coupling is switched on ($\varepsilon_{ee} = -2$); and the full LMA-Dark approximation, obtained by simultaneously transforming $\theta_{12} \rightarrow \pi/2 - \theta_{12}$ and $\varepsilon_{ee} \rightarrow -(2 + \varepsilon_{ee})$. The standard-LMA curve develops large excursions, exceeding 20% at some energies, that are visibly inconsistent with the Super-Kamiokande measurement $A_{\text{DN}} = (3.3 \pm 1.0)\%$ (Abe et al., 2016) once an Earth-matter NSI coupling of this size is present; the LMA-Dark curve, in contrast, remains broadly compatible with the measured band across most of the energy range, illustrating why this near-degenerate solution cannot be excluded by the day–night asymmetry alone and must be resolved using complementary observables.

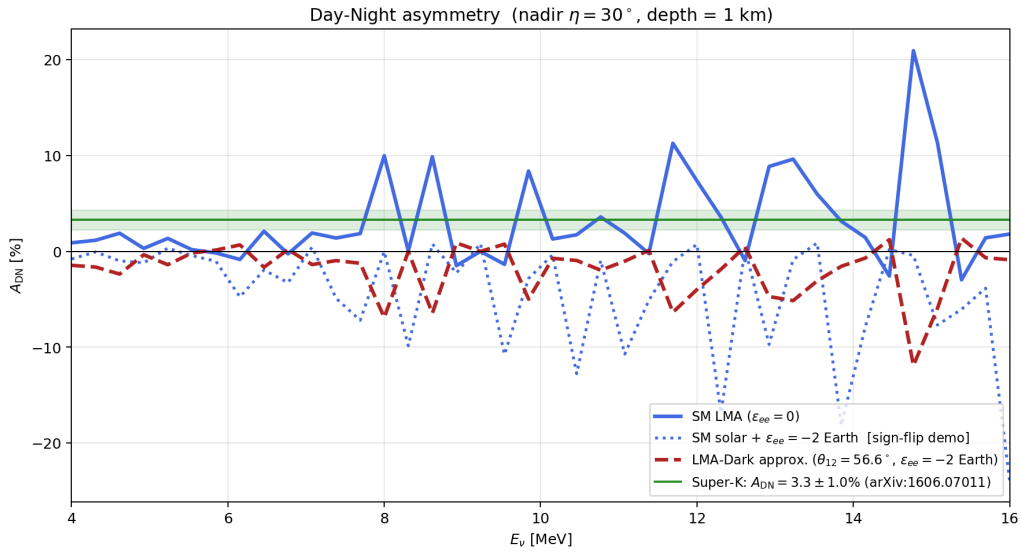


Figure 12: Day–night asymmetry versus energy at fixed nadir angle ($\eta = 30^\circ$, 1 km depth), comparing the standard LMA solution, a sign-flip demonstration with an Earth-only NSI coupling, and the full LMA-Dark approximation, against the Super-Kamiokande measurement (Abe et al., 2016).

5.5 BSM extensions

5.5.1 NSI

The propagation-NSI formalism introduced in section 2.3.1 is validated by evaluating the solar ^8B survival probability under several representative NSI presets drawn from the literature and from the projected sensitivity of upcoming experiments. Figure 13 compares the Standard Model prediction (no NSI) against the diagonal bound of Biggio et al. (2009), an off-diagonal $\varepsilon_{e\tau} = 0.15$ coupling representative of DUNE’s projected sensitivity, a combined $\varepsilon_{ee} + \varepsilon_{e\tau}$ scenario representative of Hyper-Kamiokande, a purely off-diagonal $\mu\tau$ coupling bounded by IceCube, and the global-fit values of Esteban et al. (2018). All presets converge to the same vacuum-averaged and adiabatic limits as the Standard Model at low and high energy, respectively, but distort the survival probability by up to ~ 4 percentage points in the intermediate MSW transition region (3–7 MeV); the largest distortions come from the global-fit and combined Hyper-K-like presets, while the purely $\mu\tau$ off-diagonal coupling leaves P_{ee} essentially unchanged, since it does not couple directly to the electron-flavour survival probability at this order.

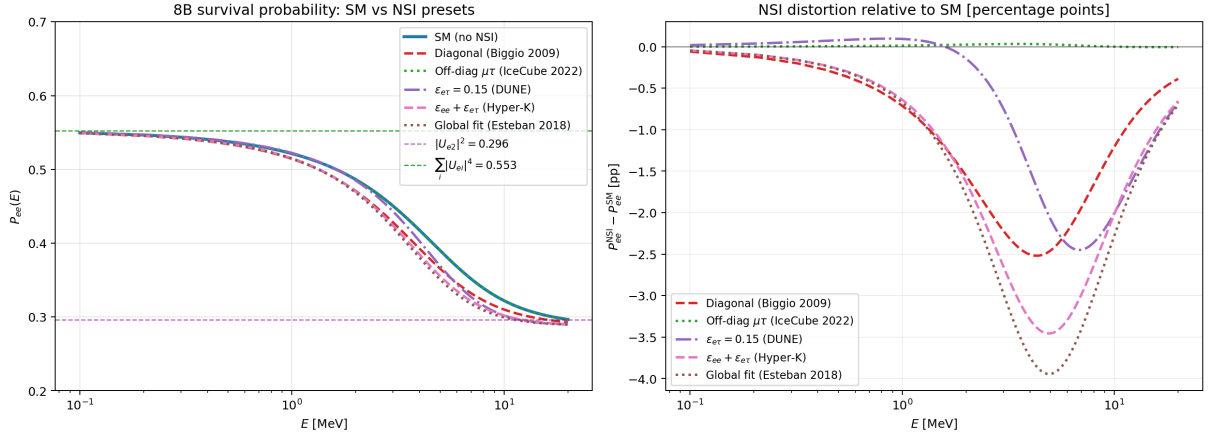


Figure 13: ^8B survival probability $P_{ee}(E)$ under the Standard Model and several representative NSI presets (left), and the resulting distortion relative to the Standard Model, in percentage points (right).

An independent scan of $\varepsilon_{e\mu}$ and $\varepsilon_{e\tau}$ over a representative range (-0.3 to $+0.3$), with all other NSI parameters at zero, produces a clear, monotonic and oppositely signed distortion of the high-energy survival probability ($E \gtrsim 3$ MeV) relative to the Standard Model for each coupling, confirming that the implementation correctly resolves both the sign and the magnitude dependence of each independent off-diagonal coupling.

5.5.2 Sterile neutrino

The $3 + 1$ sterile extension of section 2.3.1 is validated against analytic expectations for three presets (a null-mixing reference, a Giunti (2017) global-fit-motivated best fit, and an IceCube (2020) benchmark), all propagated in vacuum. Table 4 summarises the confirmed, directly recorded results: the null-mixing preset recovers the Standard-Model limit to floating-point precision in the PMNS submatrix, the Hamiltonian and the vacuum probability; unitarity, $\|U_4 U_4^\dagger - I_4\|$, holds to $\sim 10^{-16}$ for all three presets; and, under the CC-only default (section 2.3.1,

no neutron-density profile supplied), the sterile state is confirmed to receive no matter potential, both its row and column of the matter Hamiltonian vanishing identically. A dedicated CP-phase scan at LSND-like kinematics finds a maximal appearance asymmetry of only $|A_{\text{CP}}| \sim 10^{-5}$, consistent with the standard result that appreciable $3 + 1$ CP violation requires simultaneous sensitivity to two independent mass-squared splittings, not available at a single short baseline.

Table 4: Confirmed $3 + 1$ sterile-extension validation results (notebooks/validation/physics/BSM/sterile1_test.ipynb).

Check	Result
SM-limit PMNS/Hamiltonian agreement (null mixing)	exact (0 to printed precision)
Unitarity $\ U_4 U_4^\dagger - I_4\ _F$ (three presets)	$1.5\text{--}3.0 \times 10^{-16}$
Total vacuum probability $\sum_\beta P_\beta$	1.00000000, max. dev. 3.3×10^{-16}
Sterile row/column of matter Hamiltonian	exactly 0
Max. appearance CP asymmetry (LSND kinematics)	0.001% at $\delta_{14} = 108.9^\circ$

For ν_e , the two-flavour disappearance approximation is exact to 10^{-14} in the idealised limit, its realistic-mixing deviation fully explained by the naive-versus-exact $|U_{e4}|^2$ convention; for ν_μ , a genuine residual of $\mathcal{O}(10^{-2})$ survives even after this correction. Mapping the three presets onto reactor, Gallium, LSND/MiniBooNE and IceCube kinematics reproduces the well-known tension of the minimal $3 + 1$ model: combining the reactor and IceCube disappearance bounds constrains the effective appearance amplitude to $A_{e\mu}^{\text{eff}} \lesssim 0.003$, below the 0.003–0.03 required by LSND, a global tension $> 3\sigma$ consistent with the literature (Böser et al., 2020). No inference notebook in the current implementation fits real short-baseline data with the sterile extension; this is left as future work.

5.6 Detector-level case study: Daya Bay

The detector and inference layers of sections 4.1.6 and 4.1.7 are validated end to end against the real, published Daya Bay reactor-antineutrino data set: 8 antineutrino detectors (ADs) at three experimental halls, 6 reactor cores, and the 8-AD-period inverse-beta-decay (IBD) prompt-energy spectra, following the methodology of section 3.5.

Table 5: Daya Bay closure test: predicted vs. observed IBD counts at Daya Bay’s own published $(\theta_{13}, \Delta m_{31}^2)$, no fit.

	Predicted	Observed	Ratio
Total (8 detectors)	3 705 407	3 114 934	1.190
Per-detector range	—	—	1.179–1.195

Table 5 shows the closure test: evaluated at Daya Bay’s own published oscillation parameters with no free normalisation, the forward model over-predicts the observed rate by $\approx 19\%$, remarkably uniformly across all 8 detectors. A dedicated normalisation audit traced this gap through every stage of the absolute-rate chain, target mass, efficiency, cross section (independently validated to $\sim 1\%$ against Vogel & Beacom’s own quoted coefficient), and the Huber–Mueller yield, finding no unit or implementation error: part of the gap is the well-known, independently measured Reactor Antineutrino Anomaly ($\sim 5\text{--}6\%$, measured/predicted flux ratio 0.946 ± 0.020 An et al., 2017), and the remainder is consistent with the correlated normalisation systematics

Daya Bay’s own combined analysis also carries, precisely why a free ν_{norm} nuisance is included in the fits below rather than fixed to 1.

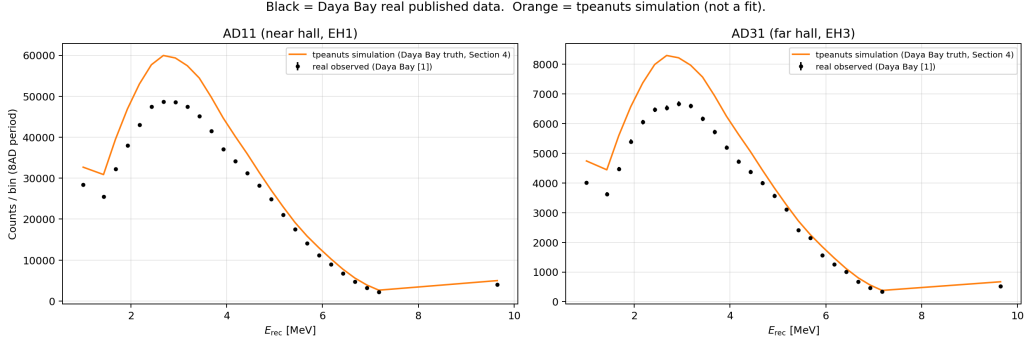


Figure 14: Predicted (`tpeanuts`, at Daya Bay’s own published truth) vs. real observed prompt-energy spectra, near-hall (AD11) and far-hall (AD31) detectors.

Figure 14 compares the predicted and observed prompt-energy spectra for one near-hall and one far-hall detector: both share the same IBD-kinematics-driven shape, with the far-hall spectrum visibly suppressed relative to its no-oscillation expectation, the qualitative signature the measurement is built on. Fitting the full 8-detector, 208-bin spectral shape jointly over $(\theta_{13}, \Delta m_{31}^2, \nu_{\text{norm}})$ was attempted with increasingly rich nuisance freedom (background-rate and energy-scale-non-linearity pulls); none of the three attempts converges to a physical point, and adding energy-scale pulls makes the fit worse, $\hat{\theta}_{13}$ collapsing to 0° , a genuine degeneracy between the oscillation deficit and uncorrelated energy-scale freedom rather than a bug. This is reported as a diagnosed limitation: it demonstrates why real reactor experiments build a full correlated systematic covariance rather than a handful of independent nuisances once sub-percent shape residuals dominate the statistical error.

Two simpler observables, by contrast, converge robustly. Integrating each detector’s spectrum to a single rate and fitting $(\theta_{13}, \nu_{\text{norm}})$ with Δm_{31}^2 fixed gives $\hat{\theta}_{13} = 8.389^\circ \pm 0.139^\circ$ ($\sin^2 2\theta_{13} = 0.08332$), -2.66% relative to Daya Bay’s published 0.08560 (-0.85σ). The near/far ratio fit, mirroring Daya Bay’s own 2012 discovery strategy (An et al., 2012), is the only strategy with genuine joint sensitivity to both parameters, and is the headline result of this case study:

Table 6: Daya Bay near/far ratio fit vs. Daya Bay’s own published best fit and NuFIT 6.1.

Quantity	<code>tpeanuts</code> (near/far)	Daya Bay published	NuFIT 6.1
$\sin^2 2\theta_{13}$	0.08442	0.0856 (An et al., 2023)	0.0878
θ_{13} [deg]	8.445 ± 0.134	8.506	8.62
Δm_{31}^2 [10^{-3}eV^2]	2.478 ± 0.055	2.528	2.511

A 2-D $-2\ln L$ scan over $(\theta_{13}, \Delta m_{31}^2)$ confirms that the fitted point and Daya Bay’s own published point sit well inside the same low- $\Delta(-2\ln L)$ region (`inference4_dayabay.ipynb`, appendix A). As table 6 shows, the near/far fit recovers both $\sin^2 2\theta_{13}$ and Δm_{31}^2 jointly, within 1–2% of Daya Bay’s own published values and comfortably inside NuFIT’s global-fit range, using `tpeanuts`’s own real multi-reactor flux, order-1/ M cross section and IAV+LSNL+Gaussian detector response, not a synthetic data set. This is the clearest demonstration that the detector and inference layers, built on top of the validated oscillation engine of section 3.2, extend

tpeanuts from fast forward simulation to genuine gradient-based analysis of real experimental data.

5.7 Other detector-inference applications

The same detector and inference layers are applied to four further real experiments (table 2), documented in full, with their own validation figures, in the specialised documents referenced in appendix A. IceCube DeepCore’s 4-parameter Poisson fit to 9 years of golden atmospheric-event data recovers $\theta_{23} = 45.42^\circ \pm 1.85^\circ$ and $\Delta m_{3l}^2 = (2.395 \pm 0.039) \times 10^{-3} \text{ eV}^2$, a -0.05σ pull on $\sin^2 \theta_{23}$ relative to the collaboration’s published best fit; a dedicated audit of the fitted muon-background normalisation (16% above the published value) rules out an Earth-matter-unitarity origin (a 0.002% effect) and instead traces most of the gap to fixed detector-systematics hypersurfaces (up to +12% for the ν_τ -CC channel) not yet freed as nuisance parameters. SNO phase II’s fit to the full 391-day, 38-observable salt-phase data set, using the real published statistical-plus-systematic covariance, converges to $\theta_{12} = 35.28^\circ \pm 1.61^\circ$, $\Delta m_{21}^2 = (6.10 \pm 3490) \times 10^{-5} \text{ eV}^2$; θ_{12} sits inside SNO’s own SNO-only 1σ range, while the reported Δm_{21}^2 uncertainty illustrates the Laplace approximation breaking down once the highly correlated systematic covariance is included, exactly why the likelihood-scan machinery of section 4.1.7 exists. Borexino’s elastic-scattering event-rate chain (target, response, cross section) is implemented and unit-tested against the real Nature 2018 low-energy spectrum (Borexino Collaboration, Agostini, et al., 2018), but its background model is currently a documented placeholder, so no fit to real data is yet reported. SNO phase I’s cross sections, response and inference model are implemented and unit-tested but not yet exercised against a full published data set.

6 Conclusions

This Master’s Thesis has presented the motivation, theoretical basis and development strategy for **tpeanuts**, a PyTorch update of PEANUTS for fast neutrino simulation. The project is based on a conservative principle: the physical conventions of PEANUTS must be preserved, while the numerical implementation is modernised to support tensor operations, batching and accelerator execution.

The main contribution is the organisation of the PEANUTS numerical core into a PyTorch-oriented design covering PMNS construction, matter potentials, vacuum propagation, matter mixing and solar-neutrino probabilities. The work also defines a validation methodology based on low-level comparisons, intermediate physical quantities and final probabilities. This methodology is essential because speed is useful only if the updated implementation remains physically faithful.

A further central achievement of this work is the modular software architecture described in sections 3.2 and 4.3, which decouples the core oscillation engine from the specific neutrino source, matter-density model and physics scenario being evaluated. This design has made it possible to extend the scope of the original PEANUTS code, limited to solar-neutrino propagation, to atmospheric-neutrino detection through the Earth; to support multiple interchangeable solar and Earth density models and alternative perturbative treatments of the matter potential; and to

incorporate beyond-the-Standard-Model extensions, namely non-standard interactions and sterile neutrinos, without modifying the underlying propagation engine. The validation and benchmark results of sections 3.2 and 3.3 confirm that these extensions integrate correctly with the rest of the framework and reproduce established physical signatures, demonstrating that the modular design is not only a software-engineering convenience but a working foundation for future physics studies.

A second, equally central contribution is the extension of this engine into a real, data-driven statistical-inference framework (sections 4.1.6 and 4.1.7): a detector-agnostic forward-folding layer and a gradient-based likelihood/fitting layer, built directly on PyTorch autograd, connect the same validated oscillation engine to real, published data from five experiments. The Daya Bay case study (section 5.6) demonstrates that this chain works end to end, a near/far ratio fit recovering both $\sin^2 2\theta_{13}$ and Δm_{31}^2 jointly within 1–2% of Daya Bay’s published values, and its diagnosed limitations (the full-spectral-shape fit’s non-convergence without a full systematic covariance, the Laplace approximation’s breakdown for SNO II’s Δm_{21}^2) identify concrete next steps rather than open questions.

Future work will focus on four lines: making the remaining detector-level analyses (full-shape reactor fit, Borexino’s background model, SNO I, IceCube’s free systematics) as robust as the Daya Bay near/far and SNO II fits already are, mainly through richer correlated systematic treatments; deepening the BSM extensions, in particular connecting the sterile extension to the real-data inference machinery already validated for the Standard-Model detector layer; generalising the propagation medium towards an arbitrary-density-profile formulation; and combining the individually validated detector-inference chains into a genuine joint likelihood across experiments.

References

- Abe, K., et al. (2016). Solar neutrino measurements in super-kamiokande-IV. *Physical Review D*, 94, 052010. <https://doi.org/10.1103/PhysRevD.94.052010>
- Aharmim, B., et al. (2005). Measurement of day and night neutrino energy spectra at SNO and constraints on neutrino mixing parameters. *Physical Review Letters*, 94, 122302. <https://doi.org/10.1103/PhysRevLett.94.122302>
- Ahmad, Q. R., et al. (2002). Direct evidence for neutrino flavor transformation from neutral-current interactions in the sudbury neutrino observatory. *Physical Review Letters*, 89, 011301. <https://doi.org/10.1103/PhysRevLett.89.011301>
- Akhmedov, E. K., Razzaque, S., & Smirnov, A. Y. (2013). Mass hierarchy, 2-3 mixing and CP-phase with huge atmospheric neutrino detectors. *Journal of High Energy Physics*, 02, 082. [https://doi.org/10.1007/JHEP02\(2013\)082](https://doi.org/10.1007/JHEP02(2013)082)
- An, F. P., et al. (2012). Observation of electron-antineutrino disappearance at daya bay. *Physical Review Letters*, 108, 171803.
- An, F. P., et al. (2017). Improved measurement of the reactor antineutrino flux and spectrum at daya bay. *Chinese Physics C*, 41, 013002.

- An, F. P., et al. (2023). Precision measurement of reactor antineutrino oscillation at kilometer-scale baselines by daya bay. *Physical Review Letters*, *130*, 161802.
- Arguelles, C. A., Salvado, J., & Weaver, C. N. (2022). nuSQuIDS: A toolbox for neutrino propagation. *Computer Physics Communications*, *277*, 108346.
- Ashie, Y., et al. (2004). Evidence for an oscillatory signature in atmospheric neutrino oscillation. *Physical Review Letters*, *93*, 101801. <https://doi.org/10.1103/PhysRevLett.93.101801>
- Baker, S., & Cousins, R. D. (1984). Clarification of the use of chi-square and likelihood functions in fits to histograms. *Nuclear Instruments and Methods*, *221*, 437.
- Biggio, C., Blennow, M., & Fernandez-Martinez, E. (2009). General bounds on non-standard neutrino interactions. *Journal of High Energy Physics*, *08*, 090.
- Borexino Collaboration, Agostini, M., et al. (2018). Comprehensive measurement of pp-chain solar neutrinos. *Nature*, *562*, 505–510.
- Böser, S., Buck, C., Giunti, C., Lesko, J., Mueller, T. A., Rasco, C., Vogel, P., & Winslow, L. (2020). Status of light sterile neutrino searches. *Progress in Particle and Nuclear Physics*, *111*, 103736. <https://doi.org/10.1016/j.pnpnp.2019.103736>
- Dziewonski, A. M., & Anderson, D. L. (1981). Preliminary reference earth model. *Physics of the Earth and Planetary Interiors*, *25*, 297–356. [https://doi.org/10.1016/0031-9201\(81\)90046-7](https://doi.org/10.1016/0031-9201(81)90046-7)
- Emmert, J. T., et al. (2021). NRLMSIS 2.0: A whole-atmosphere empirical model of temperature and neutral species densities. *Earth and Space Science*, *8*, e2020EA001321. <https://doi.org/10.1029/2020EA001321>
- Esteban, I., Gonzalez-Garcia, M. C., Maltoni, M., Martinez-Soler, I., & Schwetz, T. (2018). Updated constraints on non-standard interactions from global analysis of oscillation data. *Journal of High Energy Physics*, *08*, 180.
- Esteban, I., Gonzalez-Garcia, M. C., Maltoni, M., Schwetz, T., & Zhou, A. (2022). The fate of hints: Updated global analysis of three-flavor neutrino oscillations [NuFIT 5.2]. *Journal of High Energy Physics*, *09*, 178.
- Esteban, I., Gonzalez-Garcia, M. C., Maltoni, M., Schwetz, T., & Zhou, A. (2024). The fate of hints: Updated global analysis of three-flavor neutrino oscillations [NuFIT 6.1]. *Journal of High Energy Physics*, *12*, 216.
- Fedynitch, A., Engel, R., Gaisser, T. K., Riehn, F., & Stanev, T. (2015). Calculation of conventional and prompt lepton fluxes at very high energy. *EPJ Web of Conferences*, *99*, 08001. <https://doi.org/10.1051/epjconf/20159908001>
- Fukuda, Y., et al. (1998). Evidence for oscillation of atmospheric neutrinos. *Physical Review Letters*, *81*, 1562–1567. <https://doi.org/10.1103/PhysRevLett.81.1562>
- Giunti, C., & Kim, C. W. (2007). *Fundamentals of neutrino physics and astrophysics*. Oxford University Press.
- Gonzalez-Garcia, M. C., Halzen, F., Maltoni, M., & Tanaka, H. K. M. (2008). Radiography of the earth’s core and mantle with atmospheric neutrinos. *Physical Review Letters*, *100*, 061802. <https://doi.org/10.1103/PhysRevLett.100.061802>

- Gonzalo, T. E., & Lucente, M. (2024). PEANUTS: A software for the automatic computation of solar neutrino flux and its propagation within earth. *Computer Physics Communications*, 298, 109109. <https://doi.org/10.1016/j.cpc.2024.109109>
- Grossman, Y. (1995). Non-standard neutrino interactions and neutrino oscillation experiments. *Physics Letters B*, 359, 141.
- Haxton, W. C. (1986). Adiabatic conversion of solar neutrinos. *Physical Review Letters*, 57, 1271. <https://doi.org/10.1103/PhysRevLett.57.1271>
- Heck, D., Knapp, J., Capdevielle, J. N., Schatz, G., & Thouw, T. (1998). *CORSIKA: A monte carlo code to simulate extensive air showers* (tech. rep. No. FZKA 6019). Forschungszentrum Karlsruhe.
- Ho, J., Jain, A., & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33, 6840–6851.
- Honda, M., Kajita, T., Kasahara, K., Midorikawa, S., & Sanuki, T. (2007). Calculation of atmospheric neutrino flux using the interaction model calibrated with atmospheric muon data. *Physical Review D*, 75, 043006. <https://doi.org/10.1103/PhysRevD.75.043006>
- Huber, P. (2011). Determination of antineutrino spectra from nuclear reactors. *Physical Review C*, 84, 024617.
- Maki, Z., Nakagawa, M., & Sakata, S. (1962). Remarks on the unified model of elementary particles. *Progress of Theoretical Physics*, 28, 870–880. <https://doi.org/10.1143/PTP.28.870>
- Mikheyev, S. P., & Smirnov, A. Y. (1985). Resonance amplification of oscillations in matter and spectroscopy of solar neutrinos. *Soviet Journal of Nuclear Physics*, 42, 913.
- Mueller, T. A., et al. (2011). Improved predictions of reactor antineutrino spectra. *Physical Review C*, 83, 054615.
- Nichol, A. Q., & Dhariwal, P. (2021). Improved denoising diffusion probabilistic models. *Proceedings of Machine Learning Research*, 139, 8162–8171.
- Parke, S. (1986). Nonadiabatic level crossing in resonant neutrino oscillations. *Physical Review Letters*, 57, 1275. <https://doi.org/10.1103/PhysRevLett.57.1275>
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Kopf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., ... Chintala, S. (2019). PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems* 32.
- Pontecorvo, B. (1957). Mesonium and antimesonium. *Soviet Physics JETP*, 6, 429.
- Pontecorvo, B. (1958). Inverse beta processes and nonconservation of lepton charge. *Soviet Physics JETP*, 7, 172.
- Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 234–241.
- Sohl-Dickstein, J., Weiss, E. A., Maheswaranathan, N., & Ganguli, S. (2015). Deep unsupervised learning using nonequilibrium thermodynamics. *Proceedings of the 32nd International Conference on Machine Learning*, 37, 2256–2265.

- Song, J., Meng, C., & Ermon, S. (2021). Denoising diffusion implicit models. *International Conference on Learning Representations*.
- Tiesinga, E., Mohr, P. J., Newell, D. B., & Taylor, B. N. (2021). CODATA recommended values of the fundamental physical constants: 2018. *Reviews of Modern Physics*, 93, 025010. <https://doi.org/10.1103/RevModPhys.93.025010>
- Vinyoles, N., et al. (2017). A new generation of standard solar models. *Astrophysical Journal*, 835, 202. <https://doi.org/10.3847/1538-4357/835/2/202>
- Vogel, P., & Beacom, J. F. (1999). Angular distribution of neutron inverse beta decay. *Physical Review D*, 60, 053003.
- Wolfenstein, L. (1978). Neutrino oscillations in matter. *Physical Review D*, 17, 2369. <https://doi.org/10.1103/PhysRevD.17.2369>
- Workman, R. L., et al. (2022). Review of particle physics. *Progress of Theoretical and Experimental Physics*, 2022, 083C01. <https://doi.org/10.1093/ptep/ptac097>

A Supplementary Material

The full source code of **tpeanuts**, together with the notebooks used for validation, analysis and benchmarking, is publicly available in the following GitHub repository:

<https://github.com/juanramondiaz/tpeanuts>

The repository is organised so that the physical modules are kept separate from the notebooks that exercise them; **notebooks/** is the main entry point for reviewing results and reproducing figures (table 7).

Table 7: Notebook directory tree (paths relative to **notebooks/** in the repository above).

Path	Contents
<code>analysis/{vacuum,solar,earth,atmosphere}</code>	Fundamentals, production/exposure profiles and differential-flux analysis per medium.
<code>validation/physics/{vacuum,solarneutrino,atmosphericneutrino,BSM}</code>	Reproduction of the physical signatures of section 5.4 and table 4.
<code>validation/{nusquids,legacy}</code>	Cross-validation against nuSQuIDS and bit-level validation against legacy PEANUTS.
<code>inference/</code>	Detector-level closure tests and gradient-based fits to real data (Borexino, SNO, Daya Bay, IceCube DeepCore, SNO II; sections 5.6 and 5.7).
<code>benchmark/</code>	Performance comparison against PEANUTS and nuSQuIDS .
<code>external/{difussion,corsika,nusquids,prem,zenodo,reactor/dayabay}</code>	Generative-model training, CORSIKA generation, and exploratory/data-retrieval notebooks for external inputs.

Full architectural and physics detail beyond the scope of this thesis, module by module,

is documented separately in five companion $\text{\LaTeX}$ documents shipped alongside the code (`documentation/`), each with its own figures and, for the inference documents, its own real-data results: `guide` (`core`, `medium`, `pipeline`, `detector.common/interaction/borexino`); `BSM_NSI` and `BSM_sterile` (theory, implementation and notebook results for each BSM extension, table 4); and `inference_dayabay`, `inference_icecube`, `inference_sno_ii` (full detector implementation and real-data results underlying sections 5.6 and 5.7).